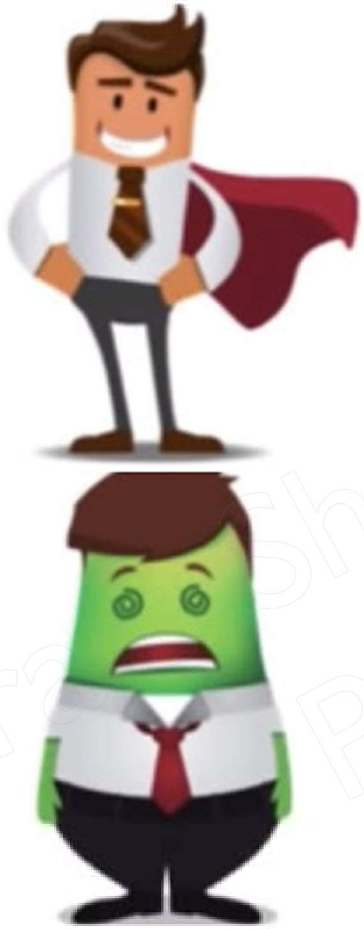
False
Positive



False
Negative

ML Testing and Error Metrics

Testing

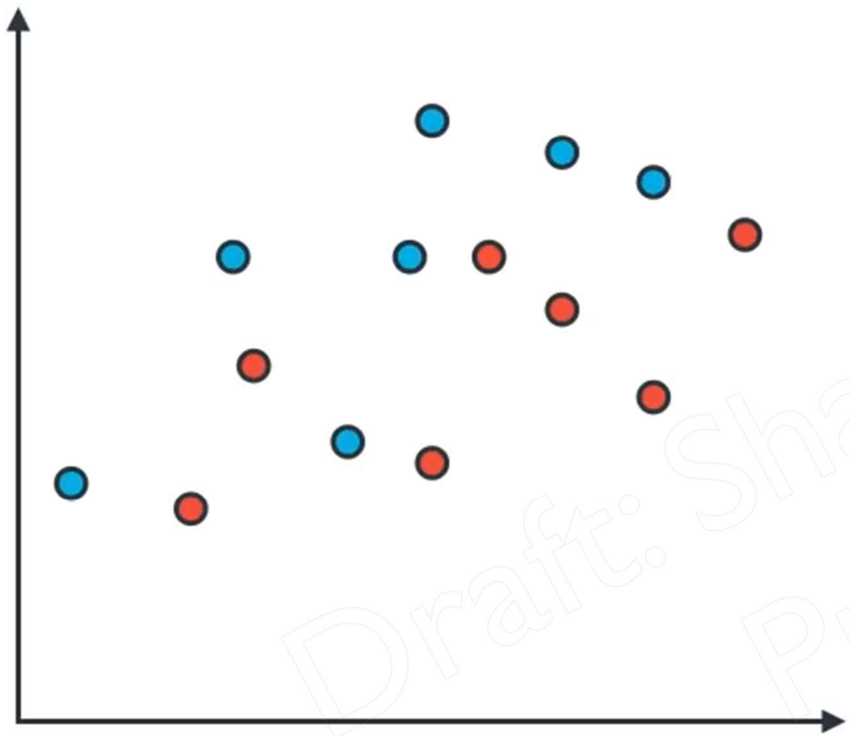
How well is my model doing?

Testing

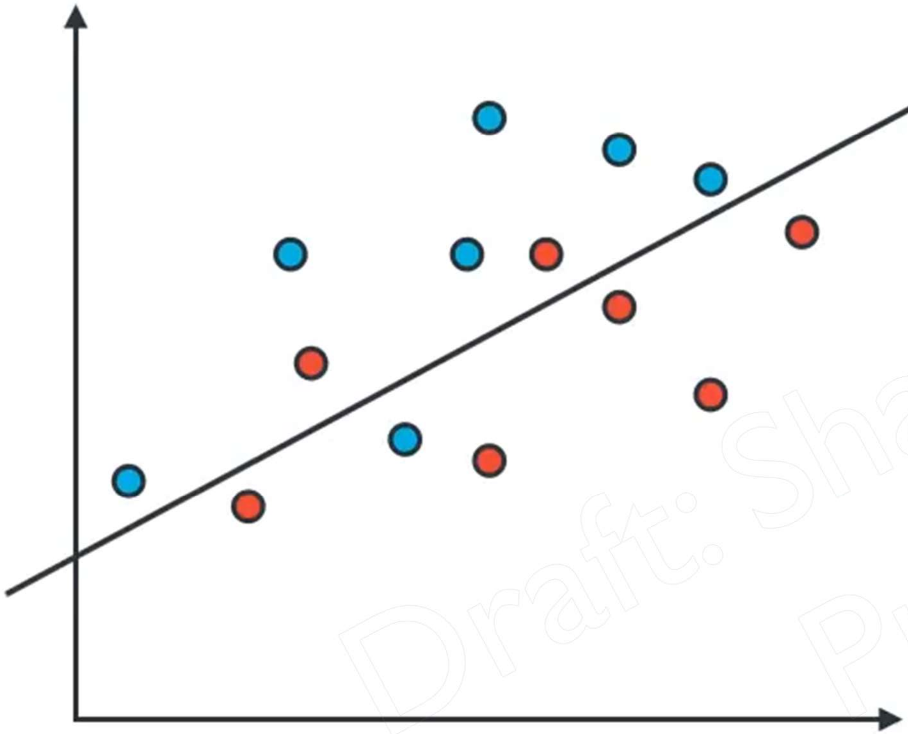
How well is my model doing?

How do I improve it?

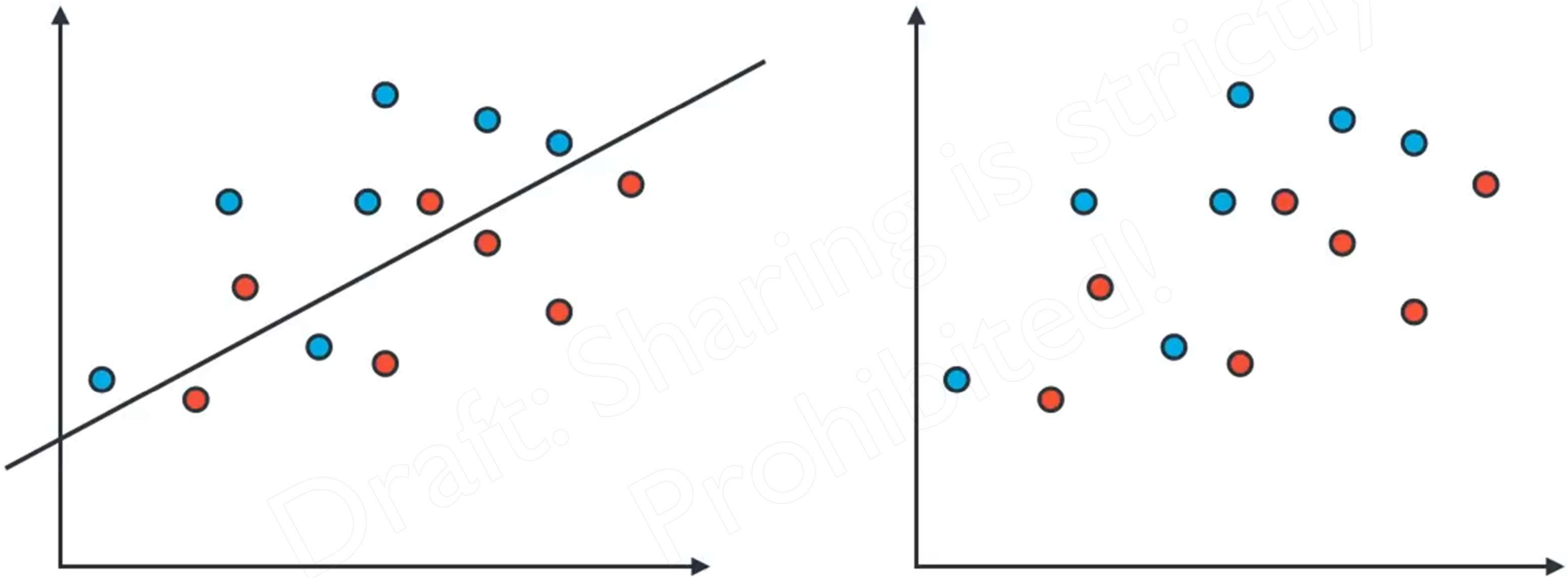
Which model is better



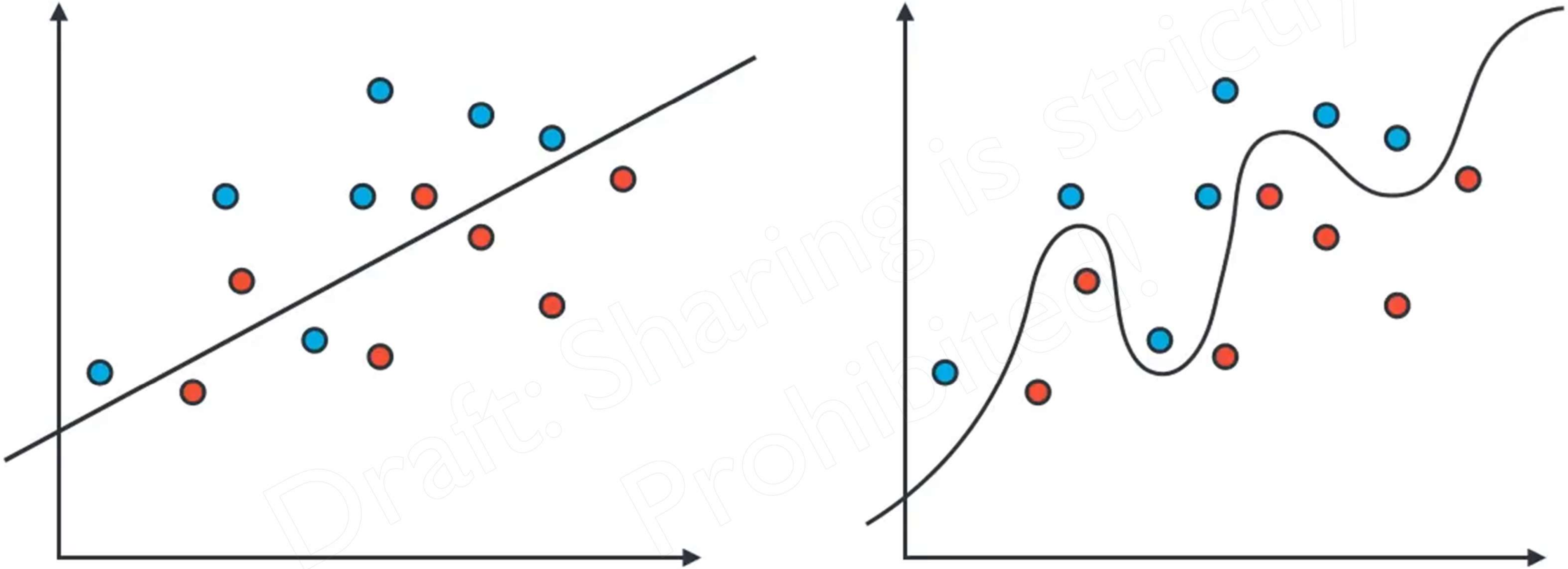
Which model is better



Which model is better



Which model is better



Evaluation Metrics

How well is my model doing?

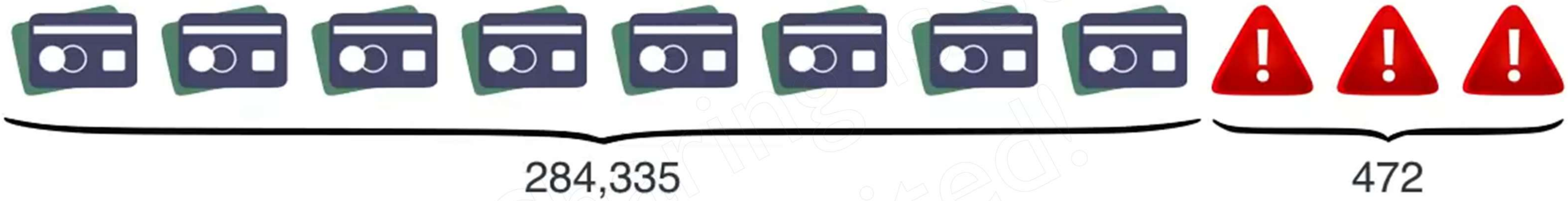
EVALUATION: LINEAR REGRESSION

Evaluation metrics for linear regression

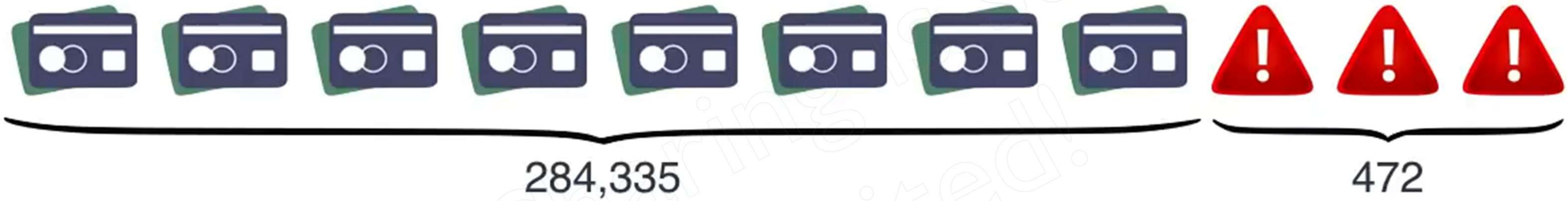
Metric	Space	Pros	Cons	When to Use
R^2	[0, 1]	Does not require comparison with other metrics to explain model fit.	Increases with the number of predictor variables, regardless of usefulness.	Simple linear regression
Adjusted R^2	[0, 1]	Adjusts the coefficient of determination for the number of predictors in the model.	The same as R^2 when there is only one predictor variable.	Multiple linear regression
MAE	≥ 0	Robust to outliers.	Does not penalise errors as extremely as other metrics.	When treating all errors equally
MSE	≥ 0	Maximises performance of linear regression.	Sensitive to outliers; magnifies large errors due to squaring.	When finding best fit models
RMSE	≥ 0	Maximises performance of linear regression.	Sensitive to outliers; magnifies large errors due to squaring.	When penalising large errors

Mean squared error	$\mathrm{MSE} = \frac{1}{n}\sum_{t=1}^{n} e_t^2$
Root mean squared error	$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n} e_t^2}$
Mean absolute error	$\mathrm{MAE} = \frac{1}{n}\sum_{t=1}^{n} \lvert e_t \rvert$
Mean absolute percentage error	$\mathrm{MAPE} = \frac{100\%}{n}\sum_{t=1}^{n} \left\lvert \frac{e_t}{y_t} \right\rvert$

Credit Card Fraud



Credit Card Fraud



Model: All transactions are good.

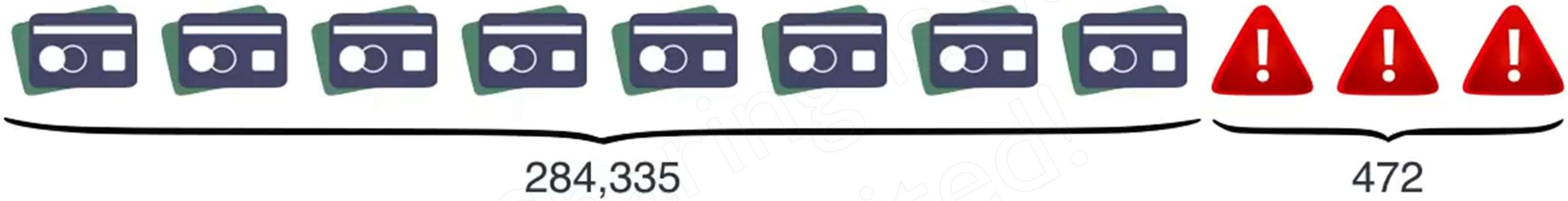
Credit Card Fraud

Model: All transactions are good.

$$\text{Correct} = \frac{284{,}335}{284{,}807} = 99.83\%$$

Problem: I'm not catching any of the bad ones!

Credit Card Fraud



Credit Card Fraud

Model: All transactions are fraudulent.

Credit Card Fraud

Model: All transactions are fraudulent.

Great! Now I'm catching *all* the bad transactions!

Credit Card Fraud

Model: All transactions are fraudulent.

Great! Now I’m catching *all* the bad transactions!

Problem: I’m accidentally catching all the good ones!

Medical Model

Case 1

Healthy

Sick

	Diagnosed Sick	Diagnosed Healthy
Sick	True positive 	False Negative 
Healthy	False Positive 	True Negative 

Confusion Matrix

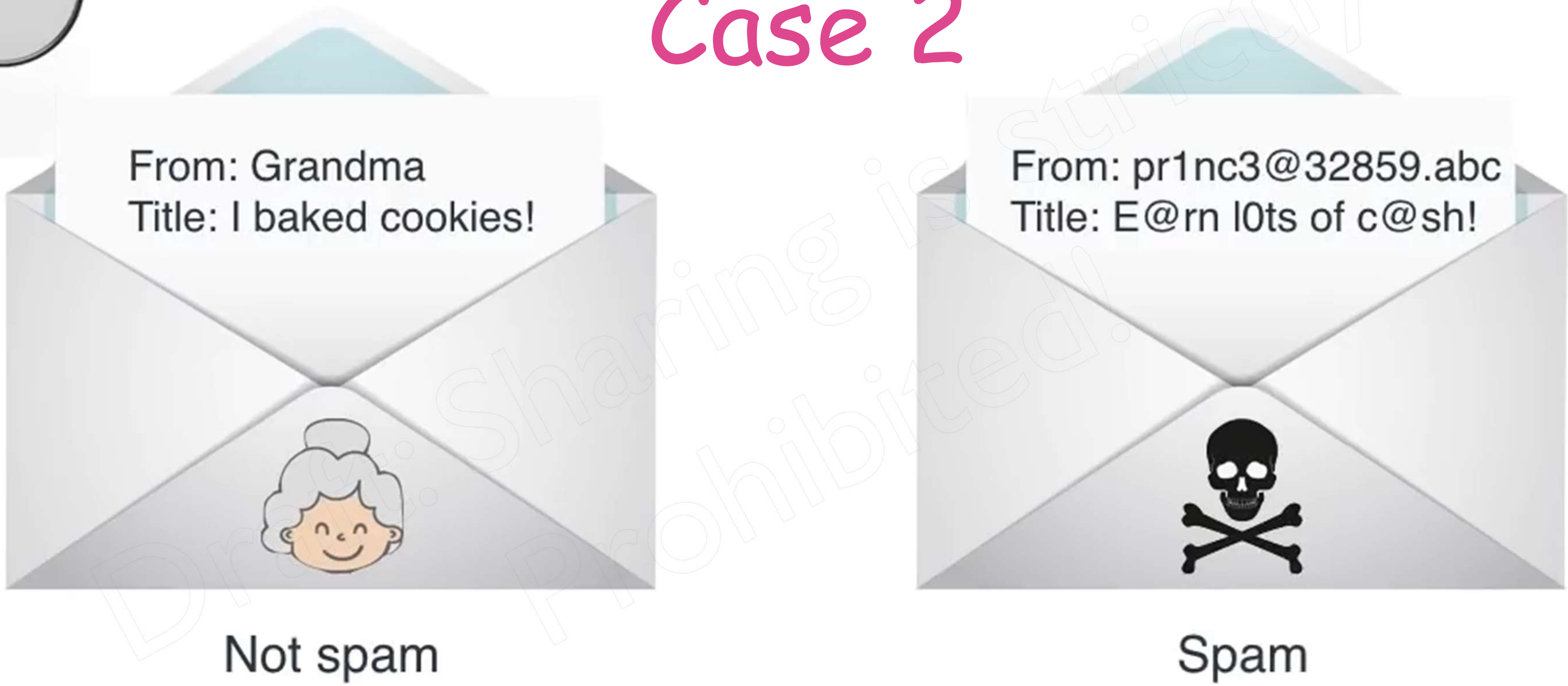
10,000
Patients

Diagnosis

Patients	Diagnosed sick	Diagnosed Healthy
Sick	1000 True positives	200 False Negatives
Healthy	800 False Positives	8000 True Negatives

Spam Classifier Model

Case 2



Confusion Matrix

1,000
e-mails

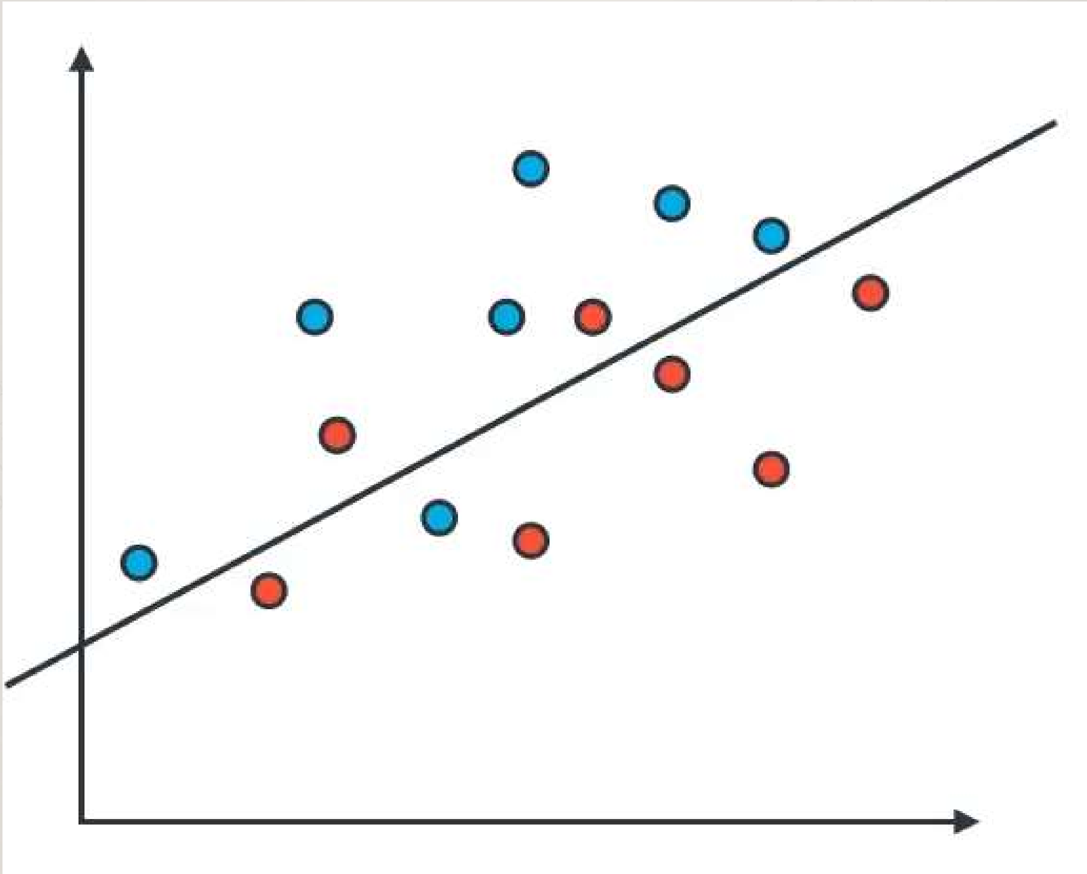
Folder

E-mail

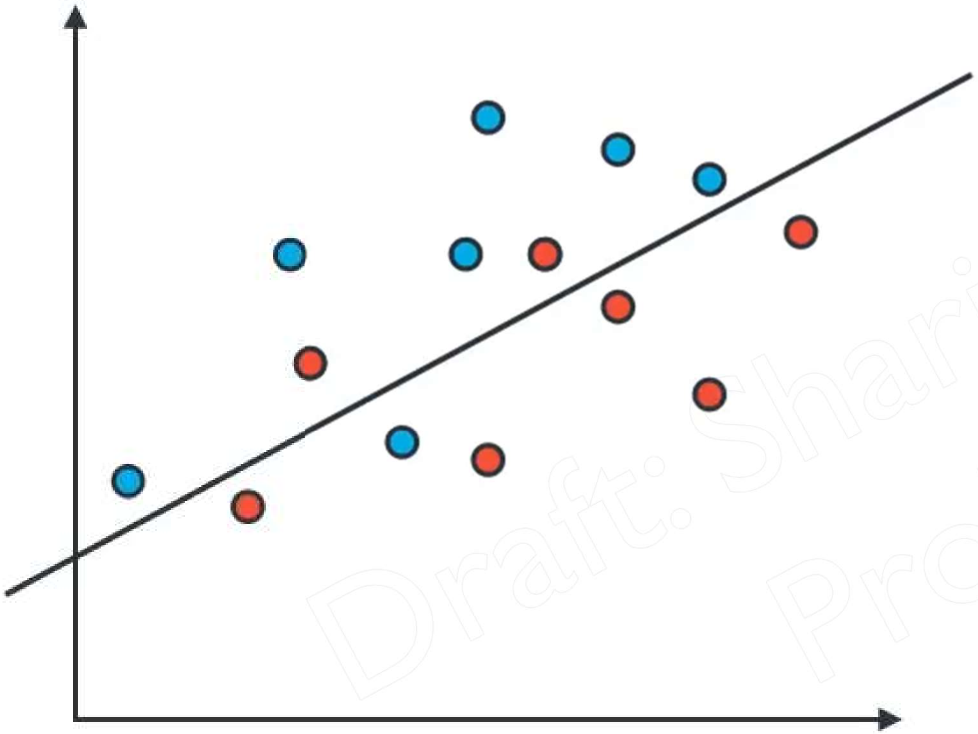
	Spam Folder	Inbox
Spam	100 True positives	170 False Negatives
Not spam	30 False Positives	700 True Negatives

Pass(positive)/Fail(Negative)

Case 3



Confusion Matrix



Data \ Prediction	Guessed Positive	Guessed Negative
Positive	6 True positives	1 False Negatives
Negative	2 False Positives	5 True Negatives

Accuracy

Diagnosis

Patients	Diagnosed sick	Diagnosed Healthy
Sick	1000	200
Healthy	800	8000

Accuracy: Out of the all the patients, how many did we classify correctly?

Accuracy

Diagnosis

Patients

	Diagnosed sick	Diagnosed Healthy
Sick	1000	200
Healthy	800	8000

Accuracy: Out of the all the patients, how many did we classify correctly?

$$\text{Accuracy} = \frac{1{,}000 + 8{,}000}{10{,}000} = 90\%$$

Accuracy

E-mail	Folder:	
	Spam Folder	Inbox
Spam	100	170
Not spam	30	700

Accuracy: Out of the all the e-mails, how many did we classify correctly?

Accuracy =

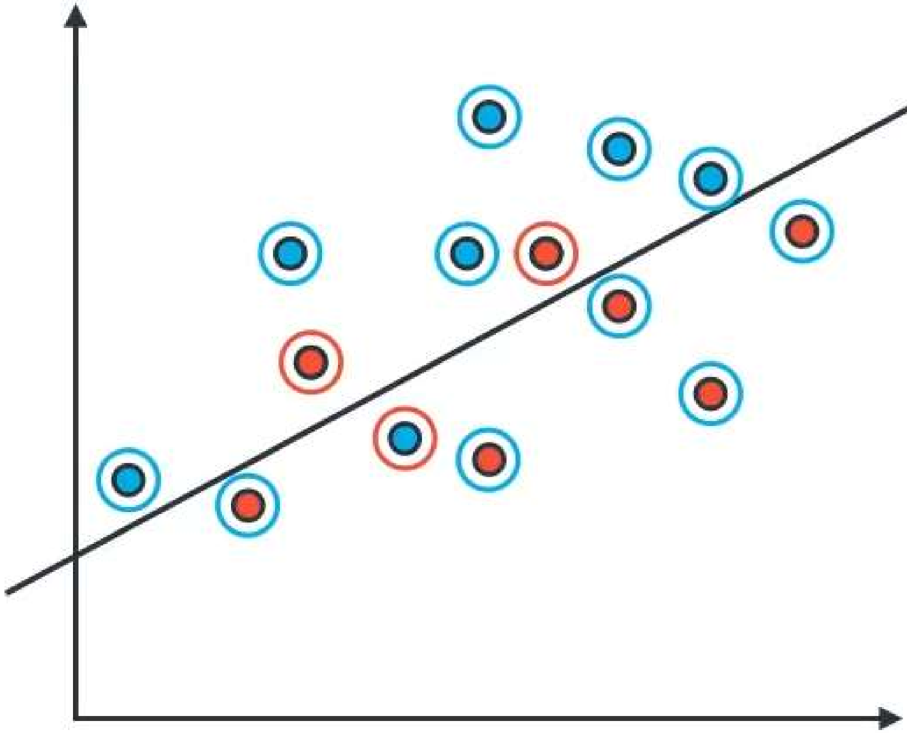
Accuracy

E-mail \ Folder	Spam Folder	Inbox
Spam	100	170
Not spam	30	700

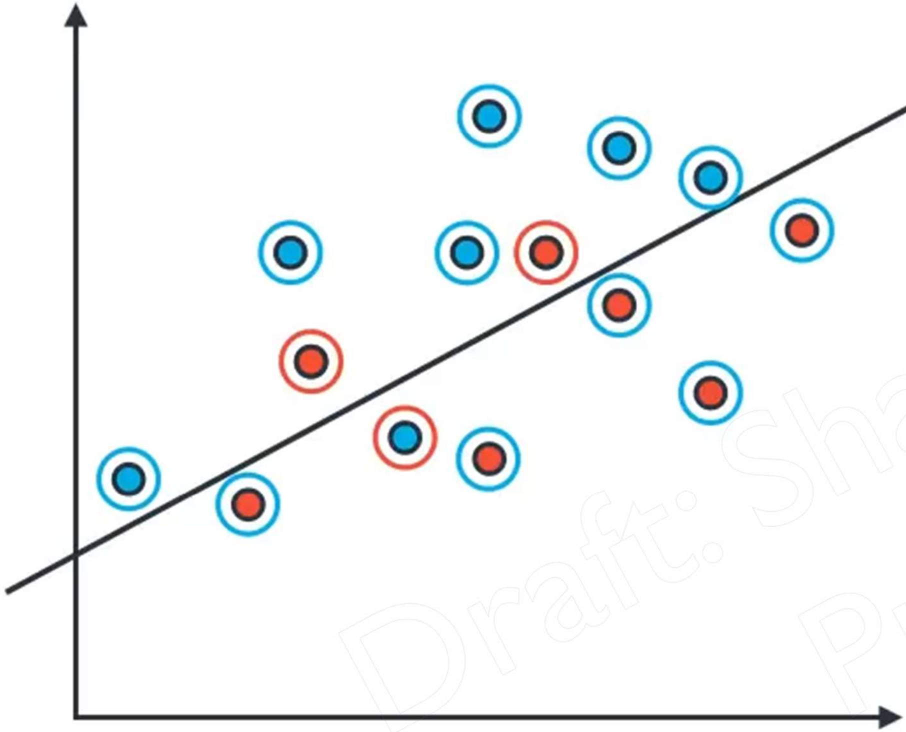
Accuracy: Out of the all the e-mails, how many did we classify correctly?

$$\text{Accuracy} = \frac{100 + 700}{1000} = 80\%$$

Accuracy



Accuracy



Precision: Out of all the data, how many points did we classify correctly?

$$\text{Accuracy} = \frac{\text{Correctly Classified points}}{\text{All points}}$$

$$= \frac{11}{11 + 3}$$

$$= \frac{11}{14}$$

$$= 78.57\%$$

EVALUATION METRICS

Medical Model

False positives ok
False negatives **NOT** ok

Spam Detector

False positives **NOT** ok
False negatives ok

EVALUATION METRICS

Medical Model

False positives ok
False negatives **NOT** ok

Find all the sick people
Ok if not all are sick

High Recall

Spam Detector

False positives **NOT** ok
False negatives ok

You don't necessarily need to find all spam
But they better all be spam

High Precision

PRECISION & RECALL

True Positive TP (++)	**False Negative FN (-)**
False Positive FP (+)	**True Negative TN (--)**

PRECISION & RECALL

True Positive TP (++)	False Negative FN (-)
False Positive FP (+)	True Negative TN (--)

Precision

PRECISION & RECALL

True Positive TP (++)	False Negative FN (-)
False Positive FP (+)	True Negative TN (--)

Recall

Precision

Diagnosis

Patients

	Diagnosed sick	Diagnosed Healthy
Sick	1000	200
Healthy	800	8000

Precision: Out of the patients
we diagnosed with an illness,
how many did we classify correctly?

$$\text{Precision} = \frac{1{,}000}{1{,}000 + 800} = 55.7\%$$

Precision

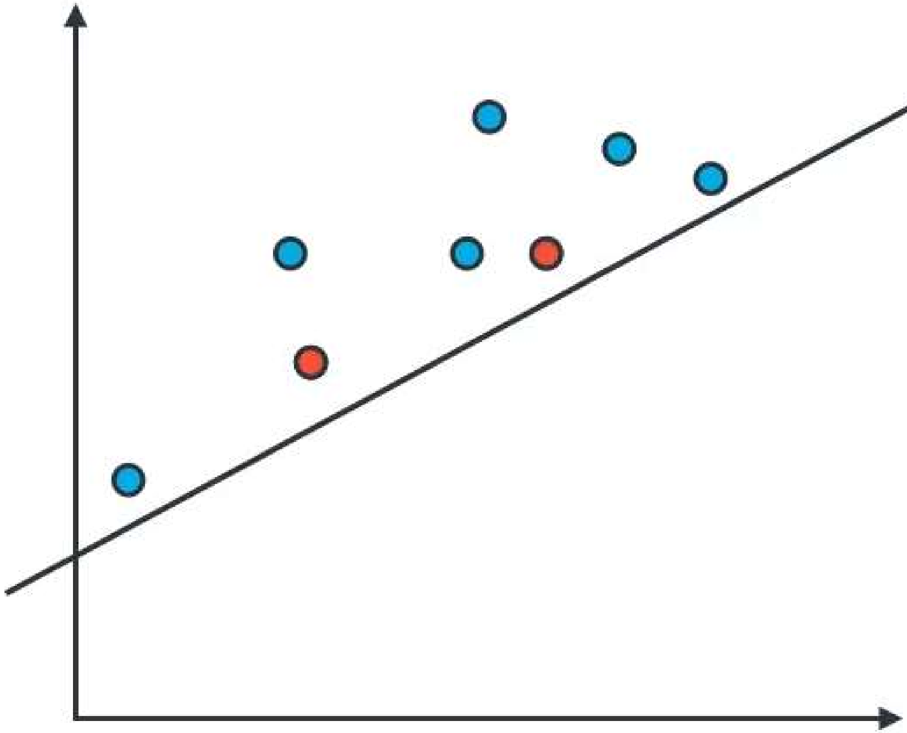
Folder (columns) / **E-mail** (rows)

	Spam Folder	Inbox
Spam	100	170
Not spam	30	700

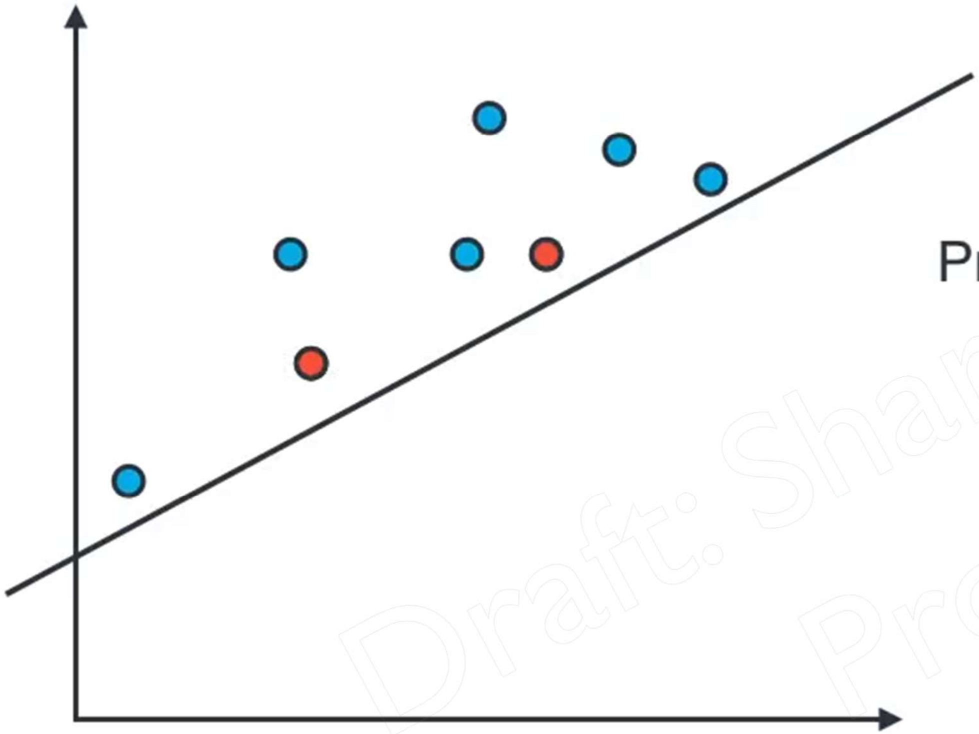
Precision: Out of the all the e-mails, sent to the spam inbox, how many were actually spam?

$$\text{Precision} = \frac{100}{100 + 30} = 76.9\%$$

Precision



Precision



Precision: Out of the points we've predicted to be positive, how many are correct?

$$\text{Precision} = \frac{\text{True positives}}{\text{True positives} + \text{False Positives}}$$

$$= \frac{6}{6 + 2}$$

$$= \frac{6}{8}$$

$$= 75\%$$

Recall

Diagnosis

Patients	Diagnosed Sick	Diagnosed Healthy
Sick	1000	200
Is Healthy	800	8000

Recall: Out of the sick patients, how many did we correctly diagnose as sick?

$$\text{Recall} = \frac{1{,}000}{1{,}000 + 200} = 83.3\%$$

Recall

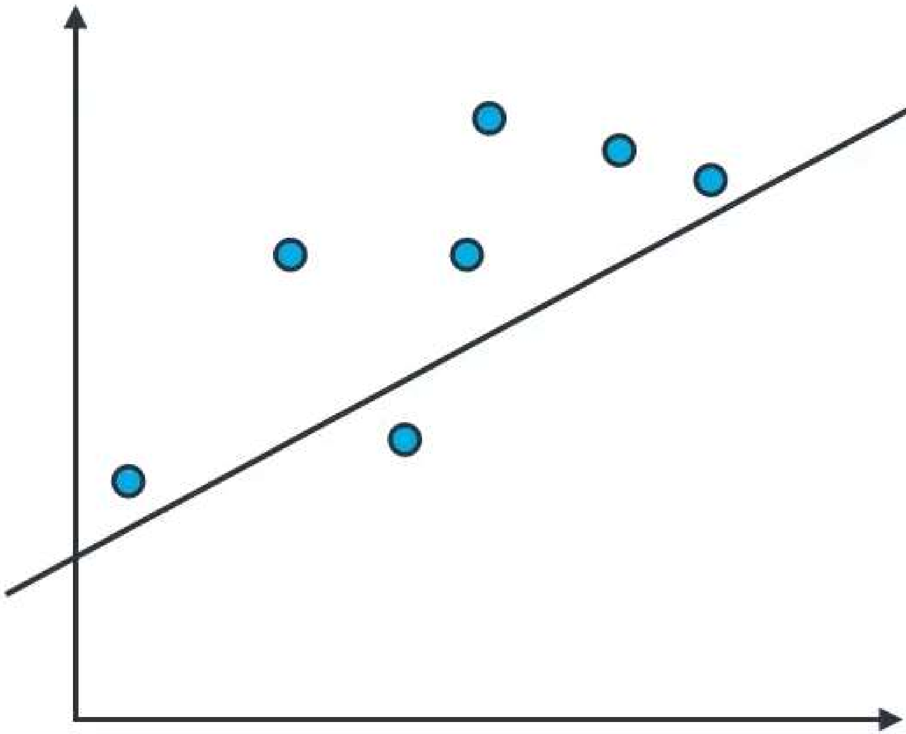
Folder

E-mail		Spam Folder	Inbox
	Spam	100	170
	Not spam	30	700

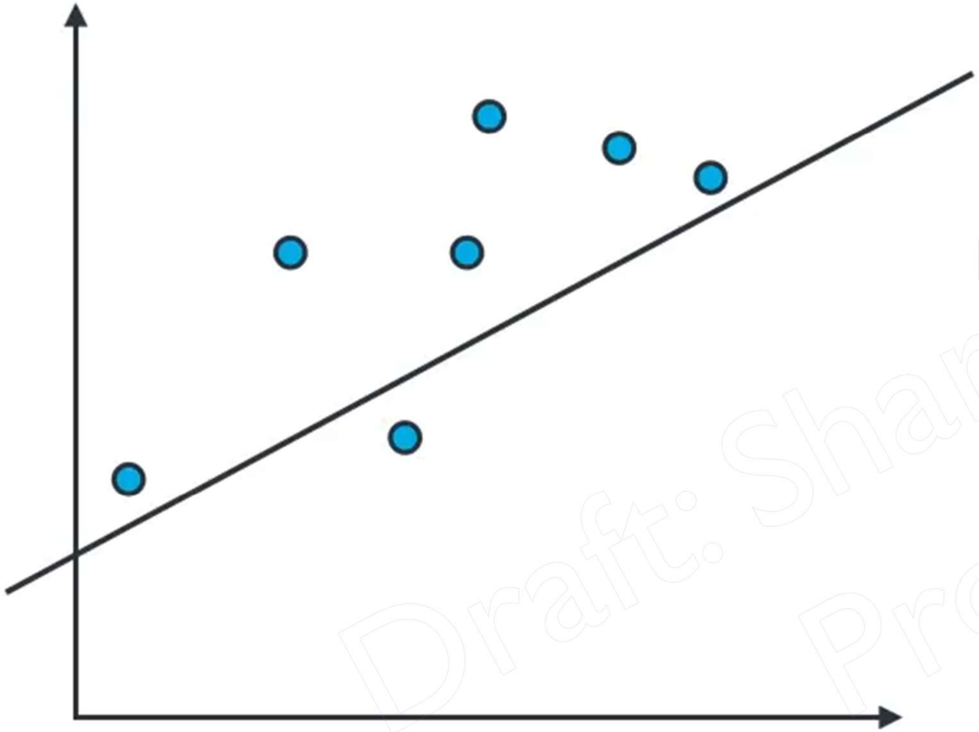
Recall: Out of the all the spam e-mails, how many were correctly sent to the spam folder?

$$\text{Recall} = \frac{100}{100 + 170} = 37\%$$

Recall



Recall



Recall: Out of the points labelled positive, how many did we correctly predict?

$$\text{Recall} = \frac{\text{True positives}}{\text{True positives} + \text{False Negatives}} = \frac{6}{6 + 1} = \frac{6}{7} = 85.7\%$$

Precision and Recall

Medical Model

Precision: 55.7%
Recall: 83.3%

Spam Detector

Precision: 76.9%
Recall: 37%

Precision and Recall

Medical Model

Precision: 55.7%
Recall: 83.3%

One score?

Spam Detector

Precision: 76.9%
Recall: 37%

Medical Model

Precision: 55.7%
Recall: 83.3%

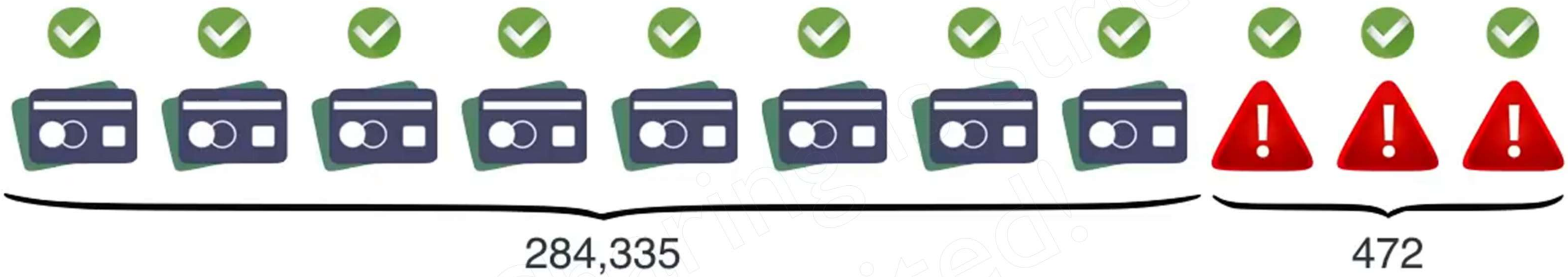
Average = 69.5%

Spam Detector

Precision: 76.9%
Recall: 37%

Average = 56.95%

Credit Card Fraud



Model: All transactions are good.

Precision = 100%

$$\text{Recall} = \frac{0}{472} = 0\%$$

Average = 50%

Credit Card Fraud

Model: All transactions are fraudulent.

$$\text{Precision} = \frac{472}{284{,}807} = .016\% \qquad \text{Recall} = \frac{472}{472} = 100\%$$

Average = 50.008%

Harmonic mean

Harmonic mean

Arithmetic Mean = $\frac{x+y}{2}$

Harmonic mean

$$\text{Arithmetic Mean} = \frac{x+y}{2}$$

$$\text{Harmonic Mean} = \frac{2xy}{x+y}$$

Harmonic mean

Precision = 1
Recall = 0
Average = 0.5
Harmonic Mean = 0

Arithmetic Mean = $\frac{x+y}{2}$

Harmonic Mean = $\frac{2xy}{x+y}$

Harmonic mean

$$\text{Arithmetic Mean} = \frac{x+y}{2}$$

$$\text{Harmonic Mean} = \frac{2xy}{x+y}$$

Precision = 1
Recall = 0
Average = 0.5
Harmonic Mean = 0

Precision = 0.2
Recall = 0.8
Average = 0.5
Harmonic Mean = 0.32

Harmonic mean

y

Arithmetic Mean = $\frac{x+y}{2}$

Harmonic Mean = $\frac{2xy}{x+y}$

x

Precision = 1
Recall = 0
Average = 0.5
Harmonic Mean = 0

Precision = 0.2
Recall = 0.8
Average = 0.5
Harmonic Mean = 0.32

~~Arithmetic Mean(Precision, Recall)~~

Harmonic mean

y

Arithmetic Mean = $\frac{x+y}{2}$

Harmonic Mean = $\frac{2xy}{x+y}$

x

Precision = 1
Recall = 0
Average = 0.5
Harmonic Mean = 0

Precision = 0.2
Recall = 0.8
Average = 0.5
Harmonic Mean = 0.32

~~Arithmetic Mean(Precision, Recall)~~
F1 Score = Harmonic Mean(Precision, Recall)

F1 Score

Medical Model

Precision = 55.7%

Recall = 83.3%

Average = 69.5%

F1 Score

Medical Model

Precision = 55.7%

Recall = 83.3%

Average = 69.5%

$$\text{F1 Score} = \frac{2 \times 55.7 \times 83.3}{55.7 + 83.3} = 66.76\%$$

F1 Score

Spam Detector
Model

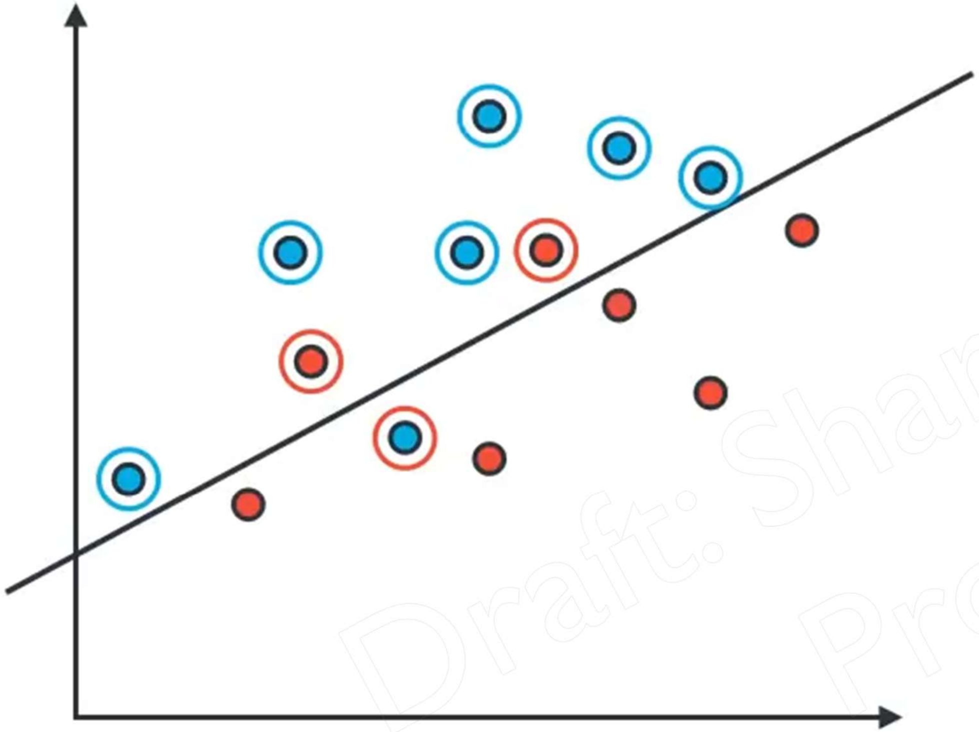
Precision = 76.9%

Recall = 37%

Average = 56.95%

$$\text{F1 Score} = \frac{2 \times 76.9 \times 37}{76.9 + 37} = 49.96\%$$

F1 Score



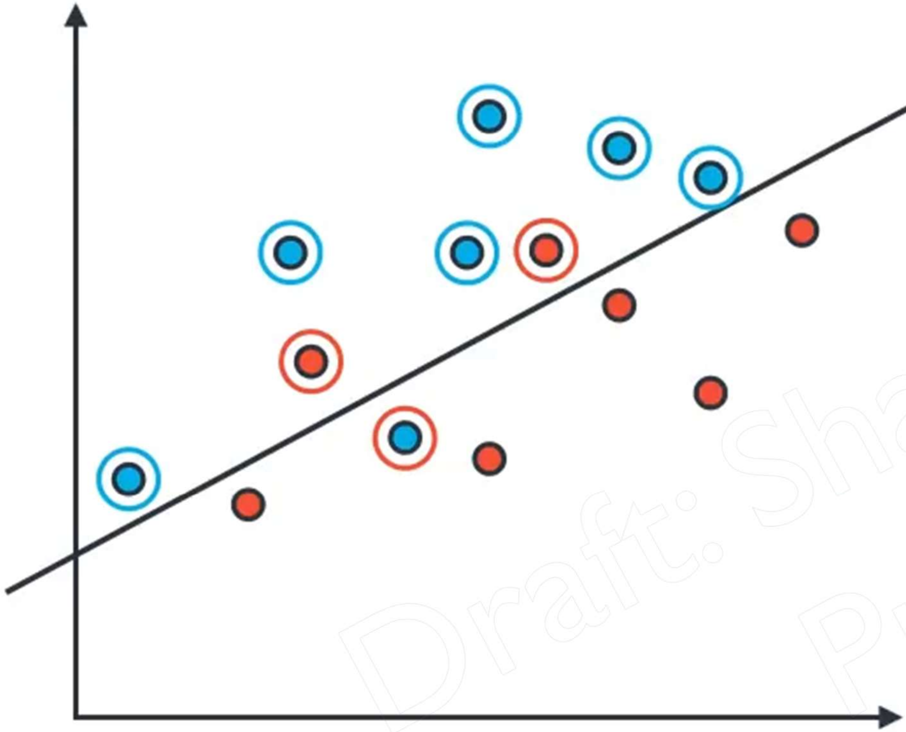
Precision = 75%

Recall = 85.7%

Average = 80.35

F1 Score = ?

F1 Score



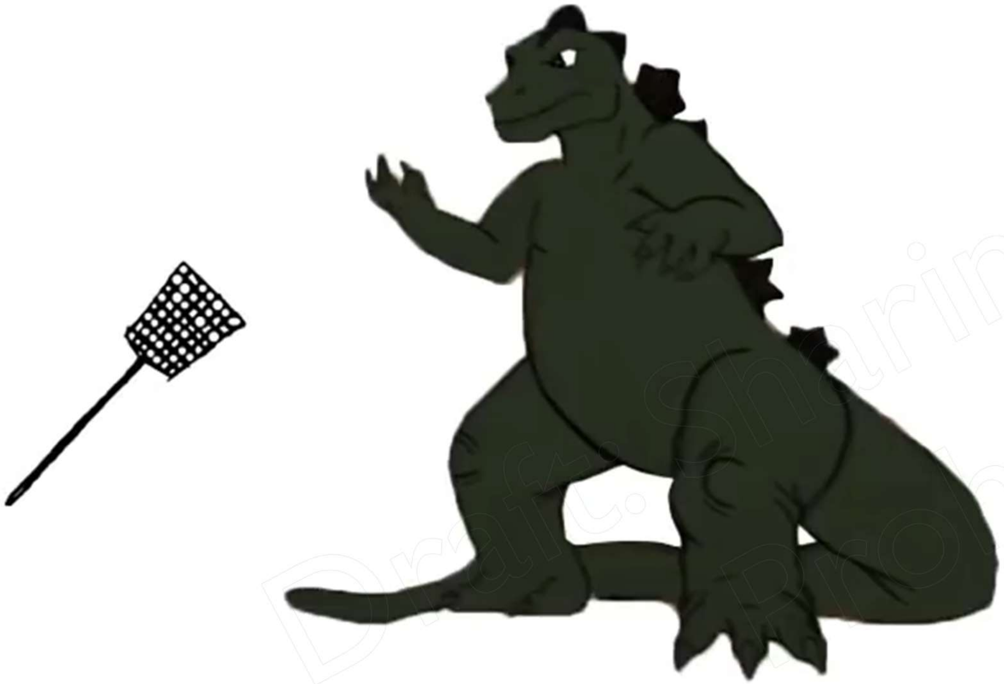
Precision = 75%

Recall = 85.7%

Average = 80.35

$$\text{F1 Score} = \frac{2 \times 75 \times 85.7}{75 + 85.7} = 80\%$$

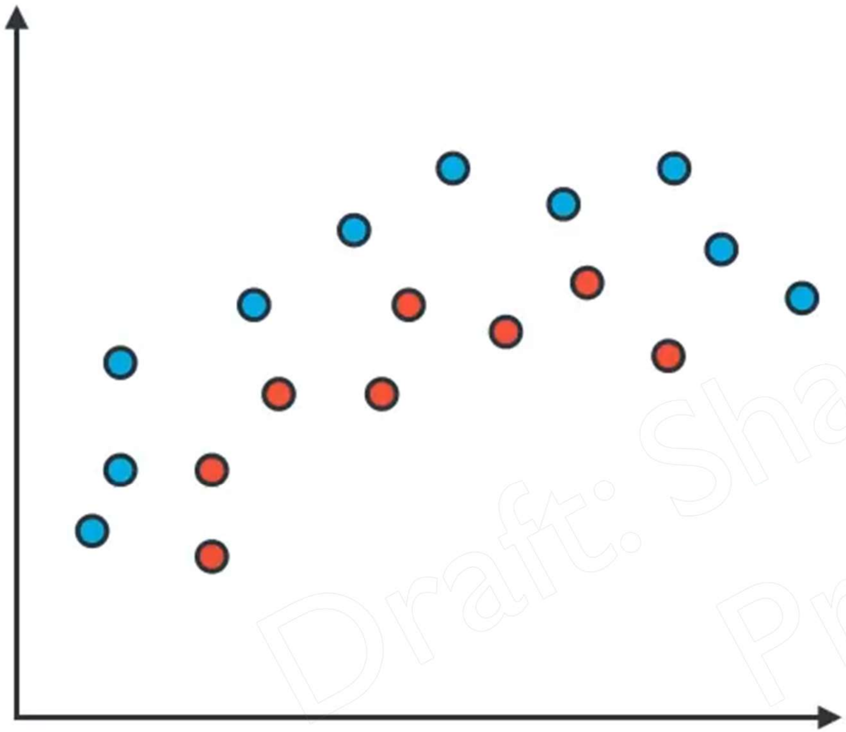
Types of Errors

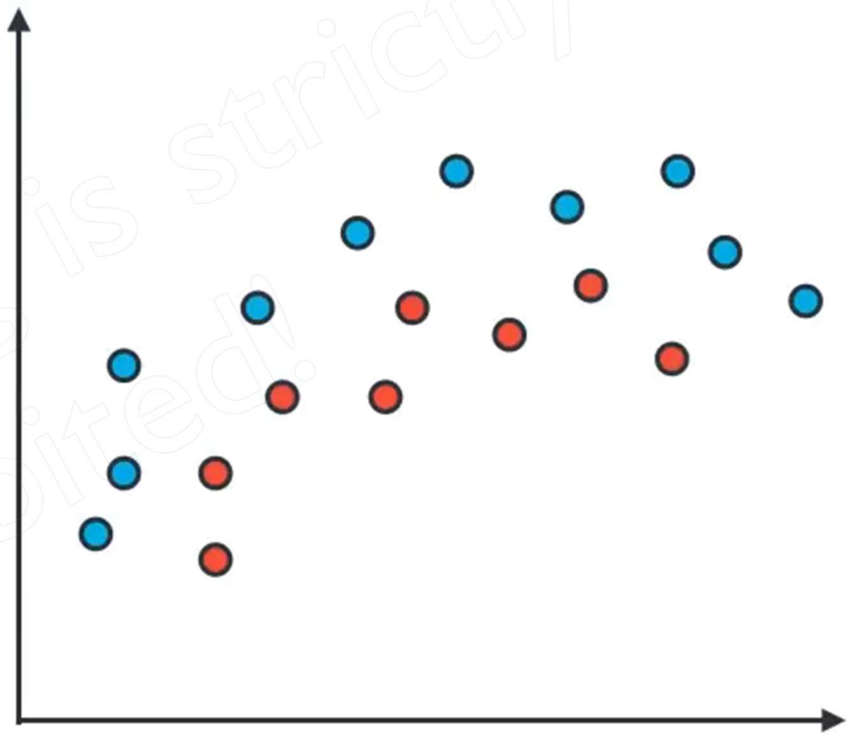


Underfitting

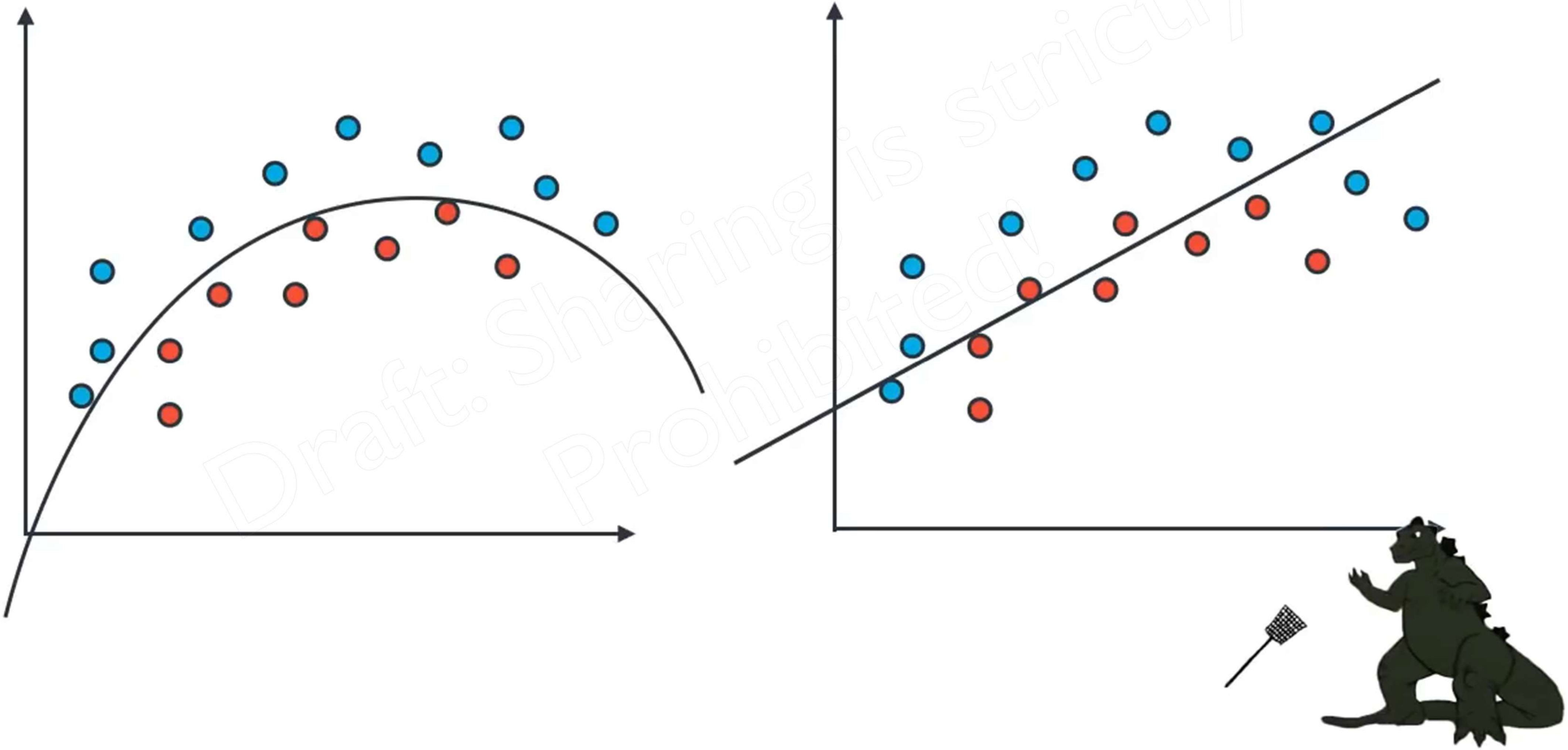
Overfitting

Error due to bias (underfitting)

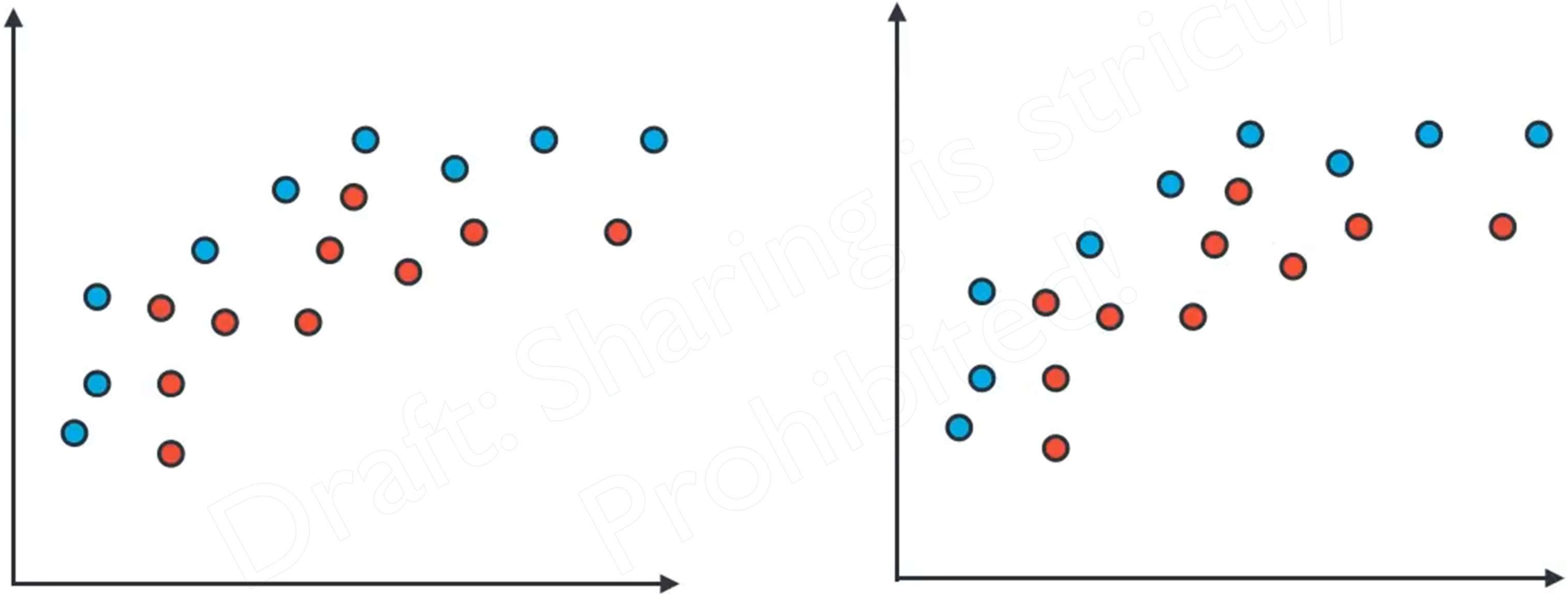




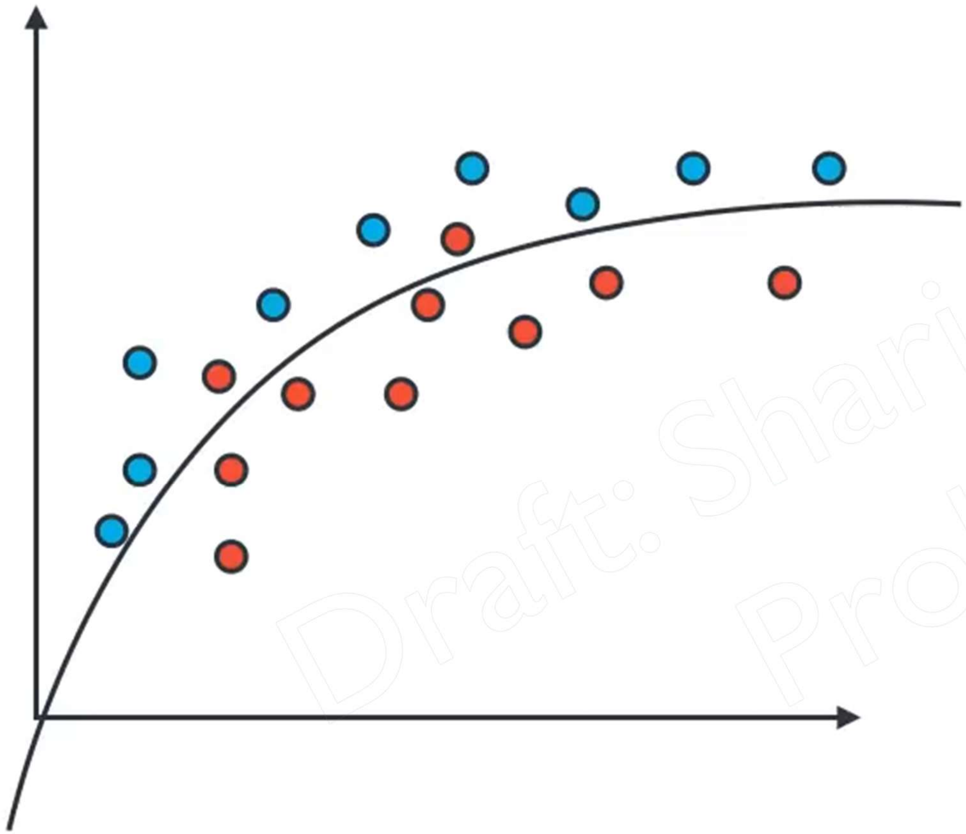
Error due to bias (underfitting)

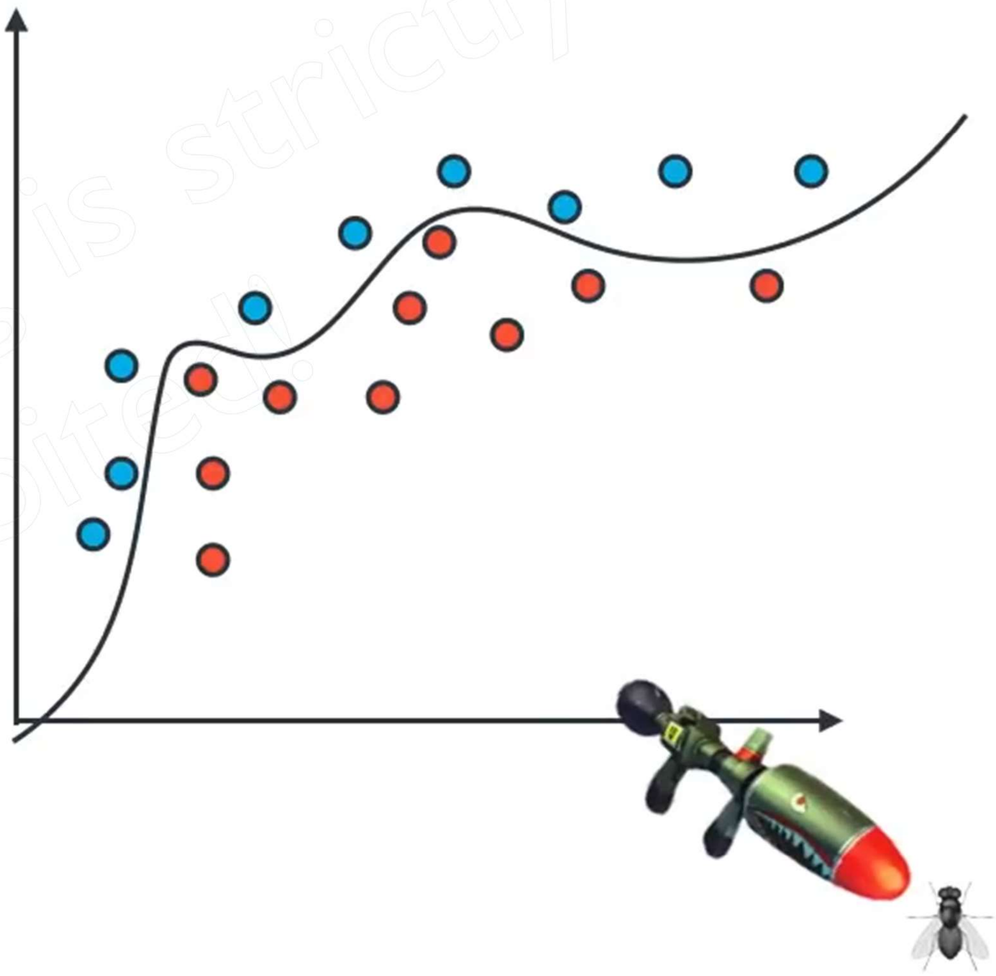


Error due to variance (overfitting)

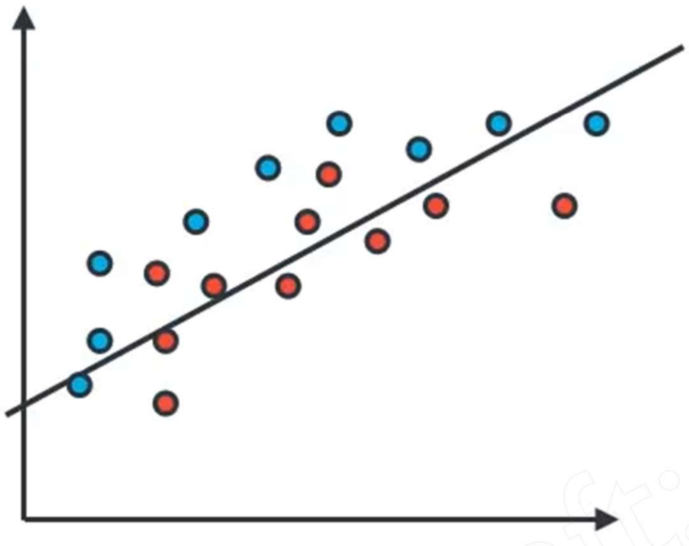


Error due to variance (overfitting)



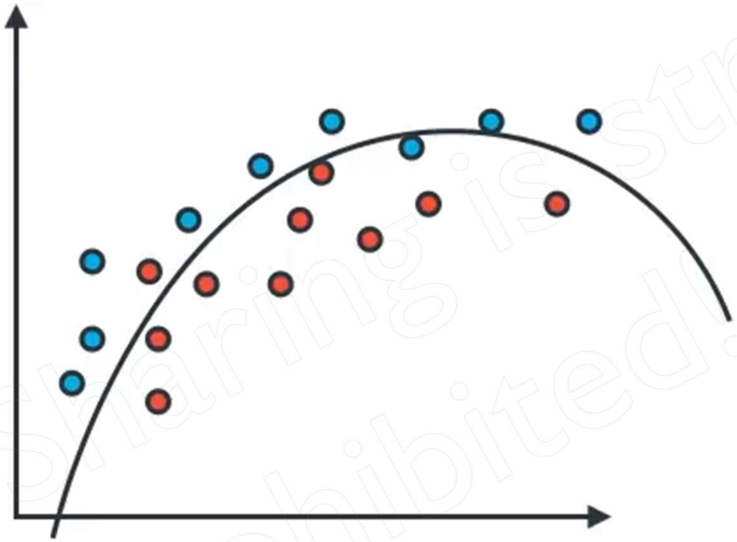


Model Complexity Graph



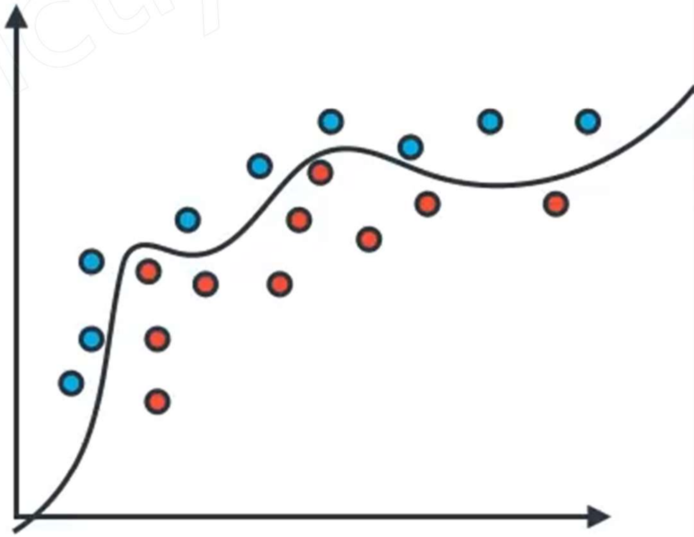
High Bias

degree = 1



Just Right

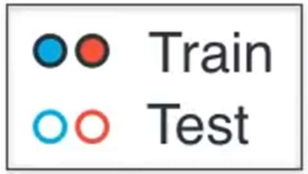
degree = 2

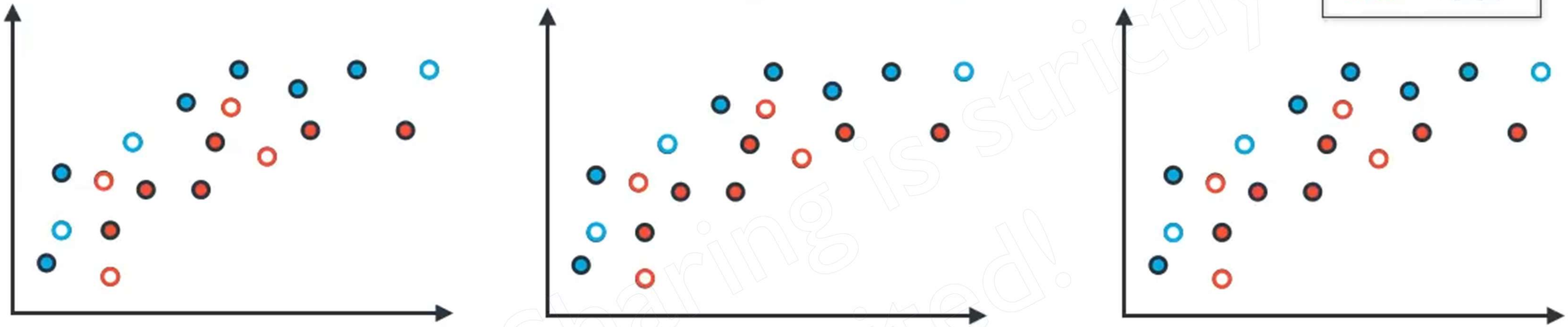


High Variance

degree = 6

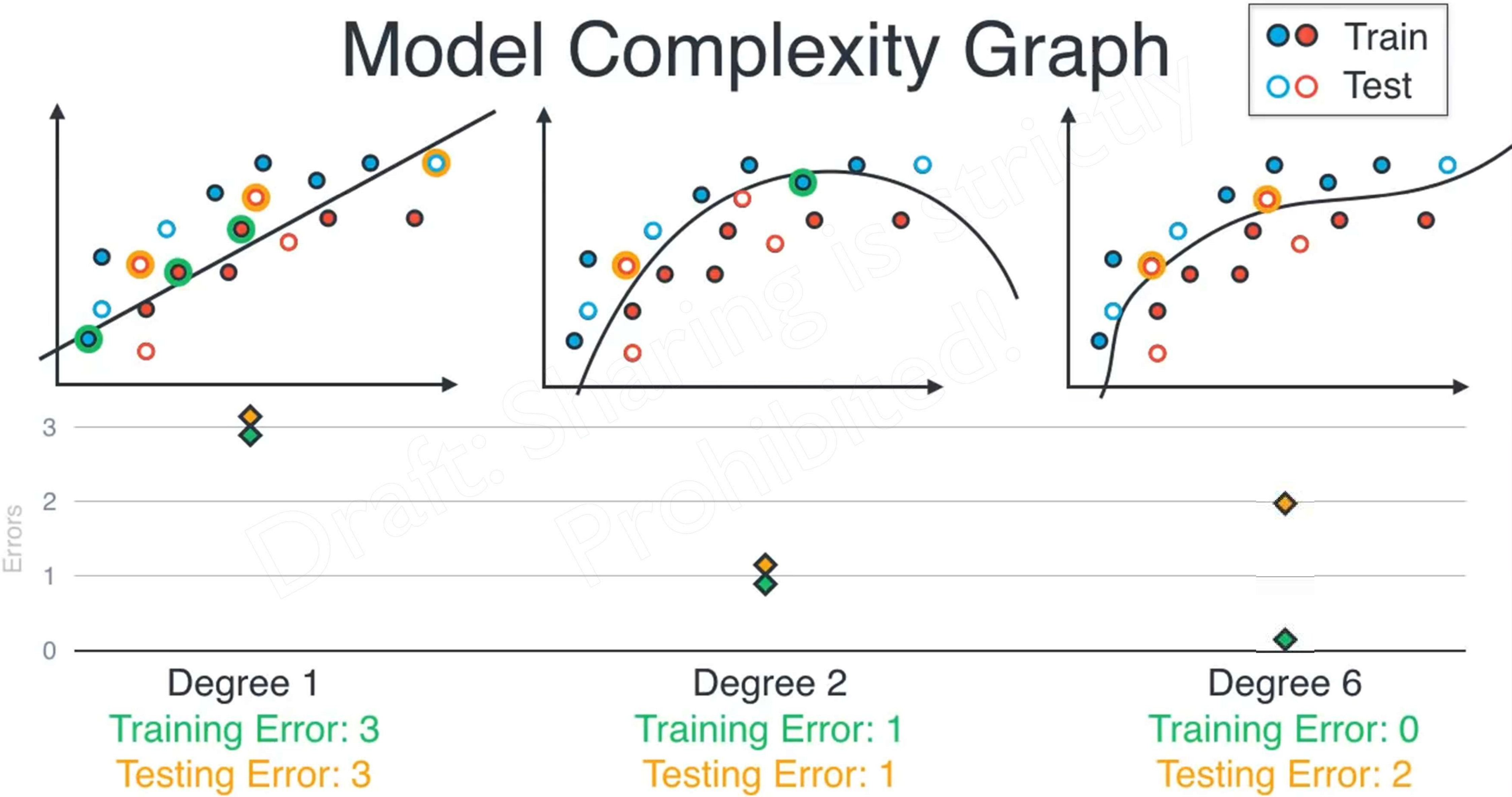
Model Complexity Graph

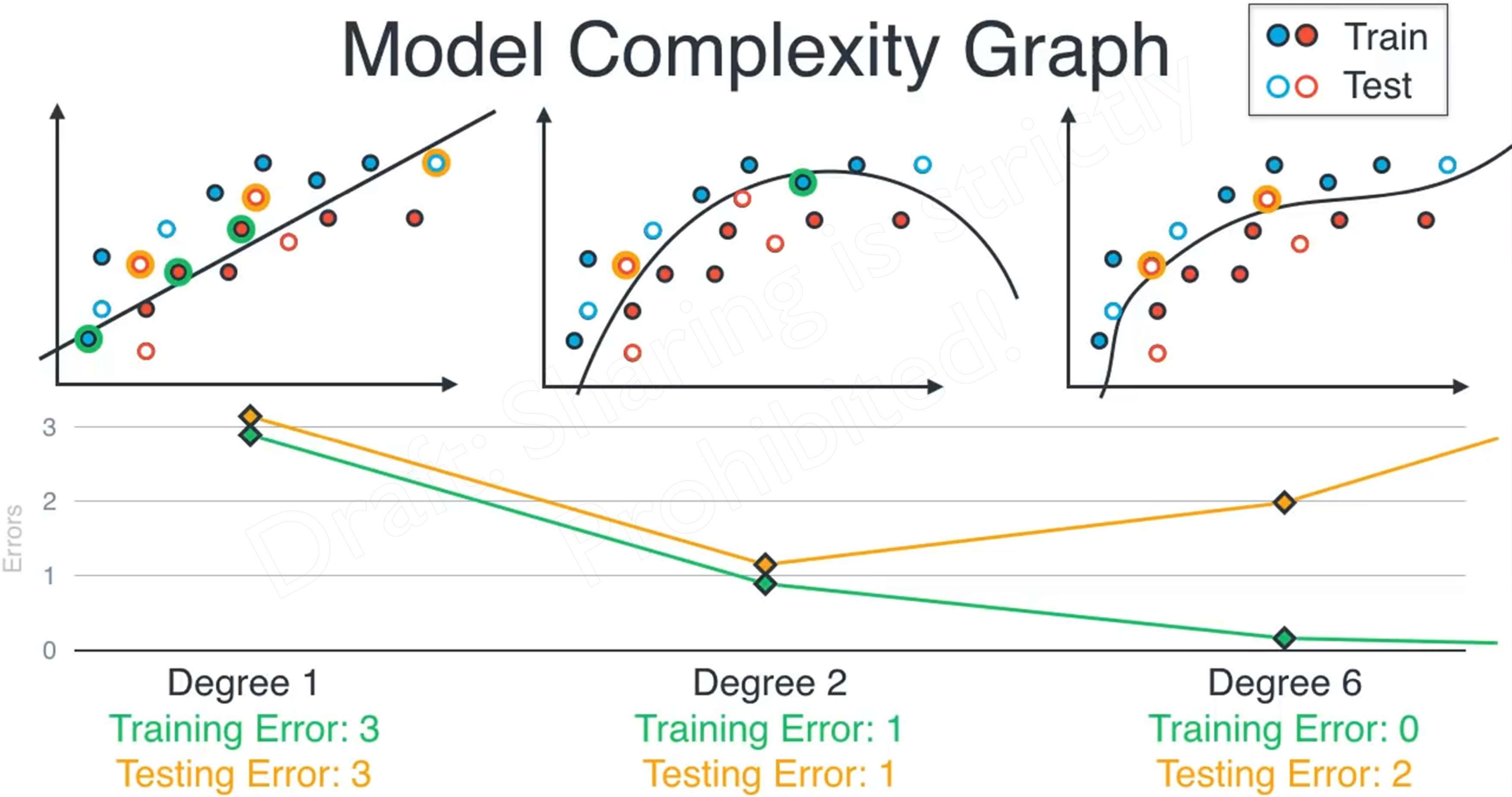


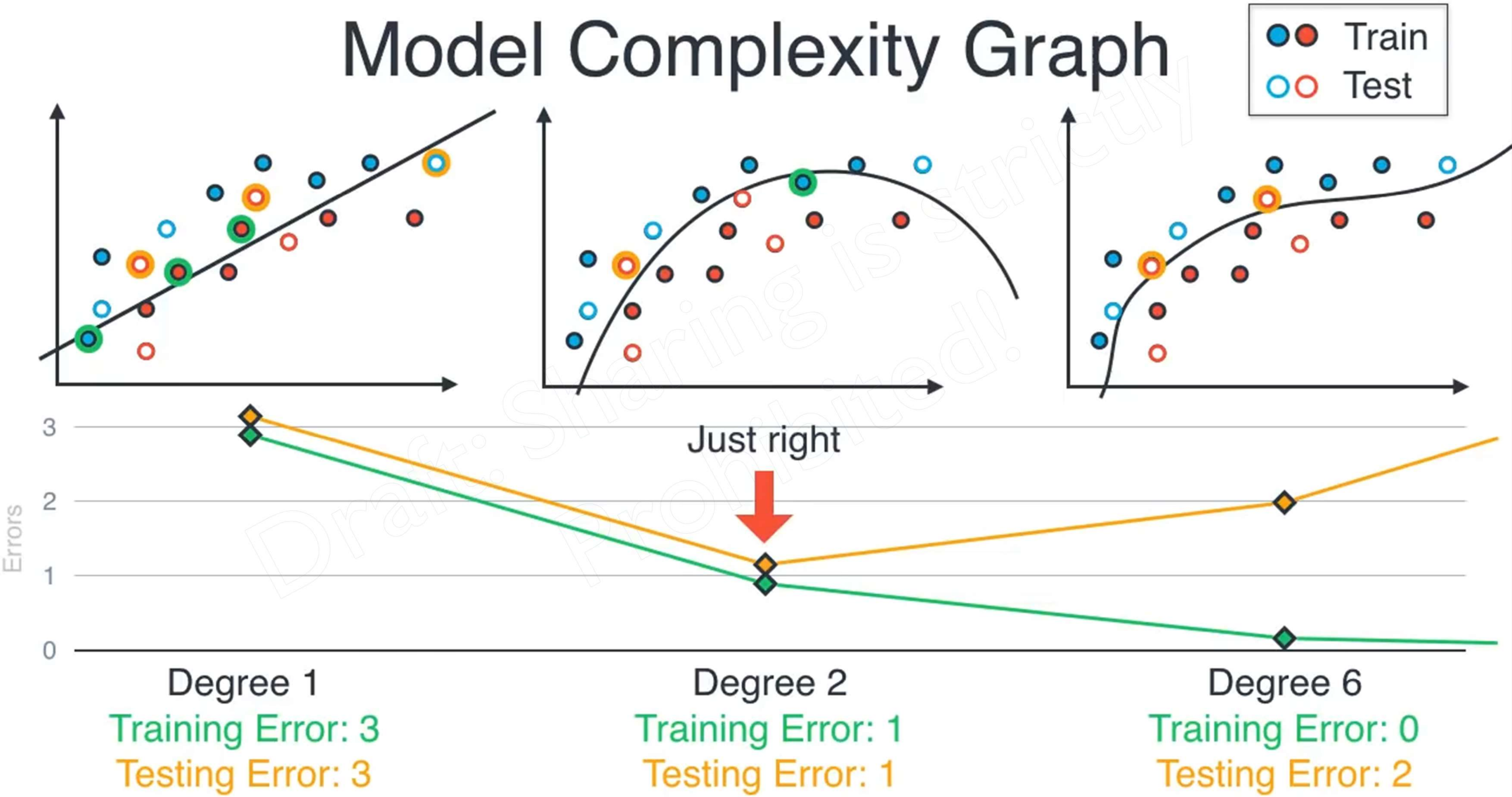


Degree 1 Degree 2 Degree 6

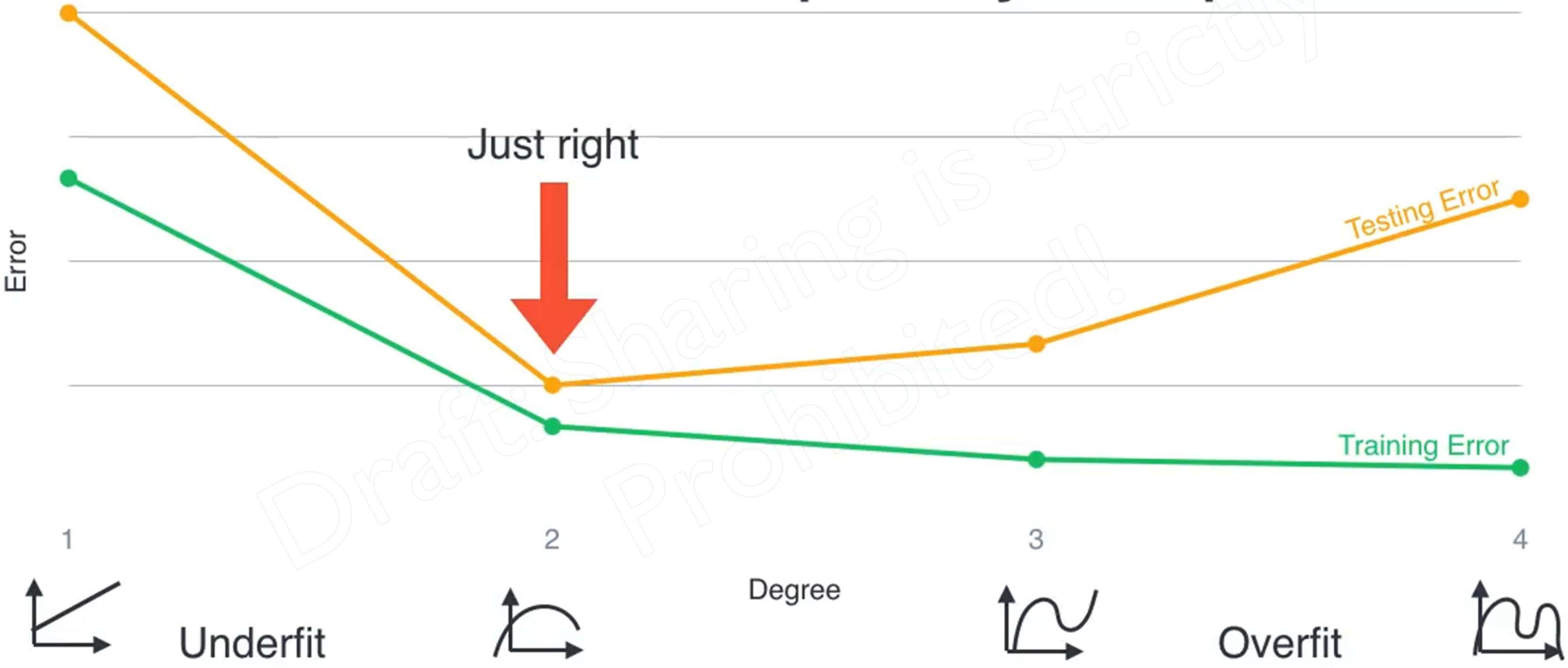
underfitting e training error o beshi abr testing error beshi.
overfitting e training error na thakai testing error beshi hbe.
tai 2 tar majhamajhi fit koratai suitble.




Model Complexity Graph
Train
Test
Errors
3
2
1
0
Degree 1
Training Error: 3
Testing Error: 3
Degree 2
Training Error: 1
Testing Error: 1
Degree 6
Training Error: 0
Testing Error: 2


Model Complexity Graph
Train
Test
Just right
3
2
1
0
Errors
Degree 1
Training Error: 3
Testing Error: 3
Degree 2
Training Error: 1
Testing Error: 1
Degree 6
Training Error: 0
Testing Error: 2

Model Complexity Graph



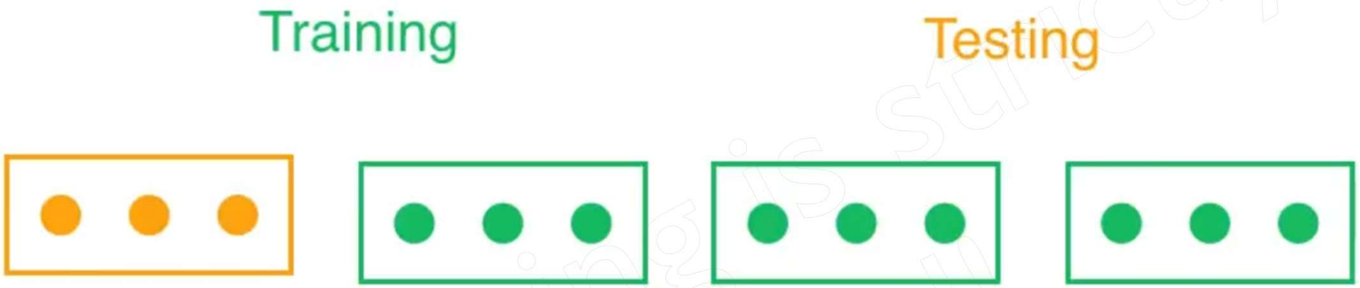
How do we not 'lose' the training data?

K-Fold Cross Validation

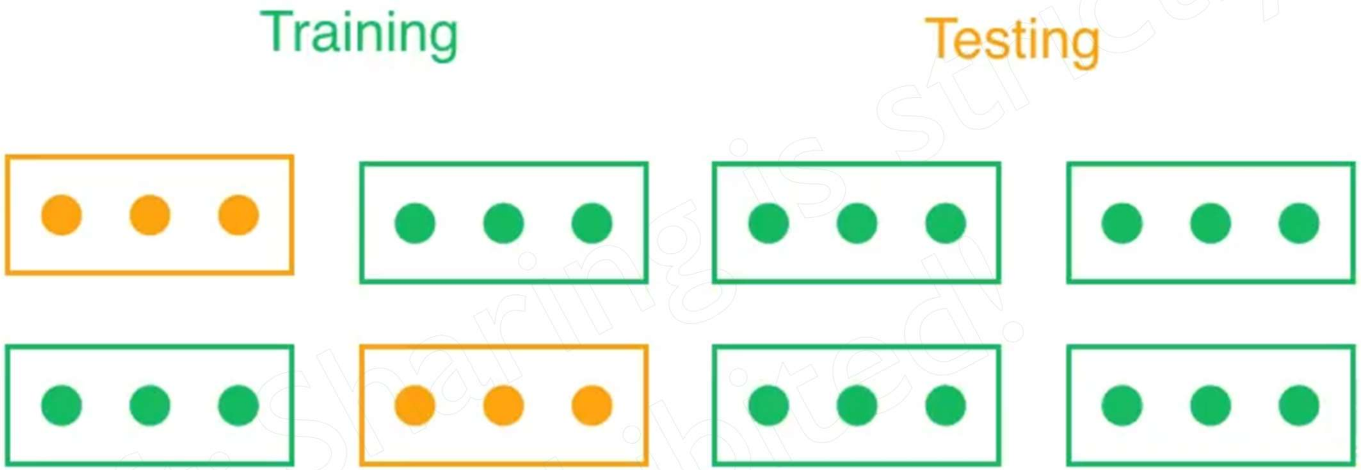
Training

Testing

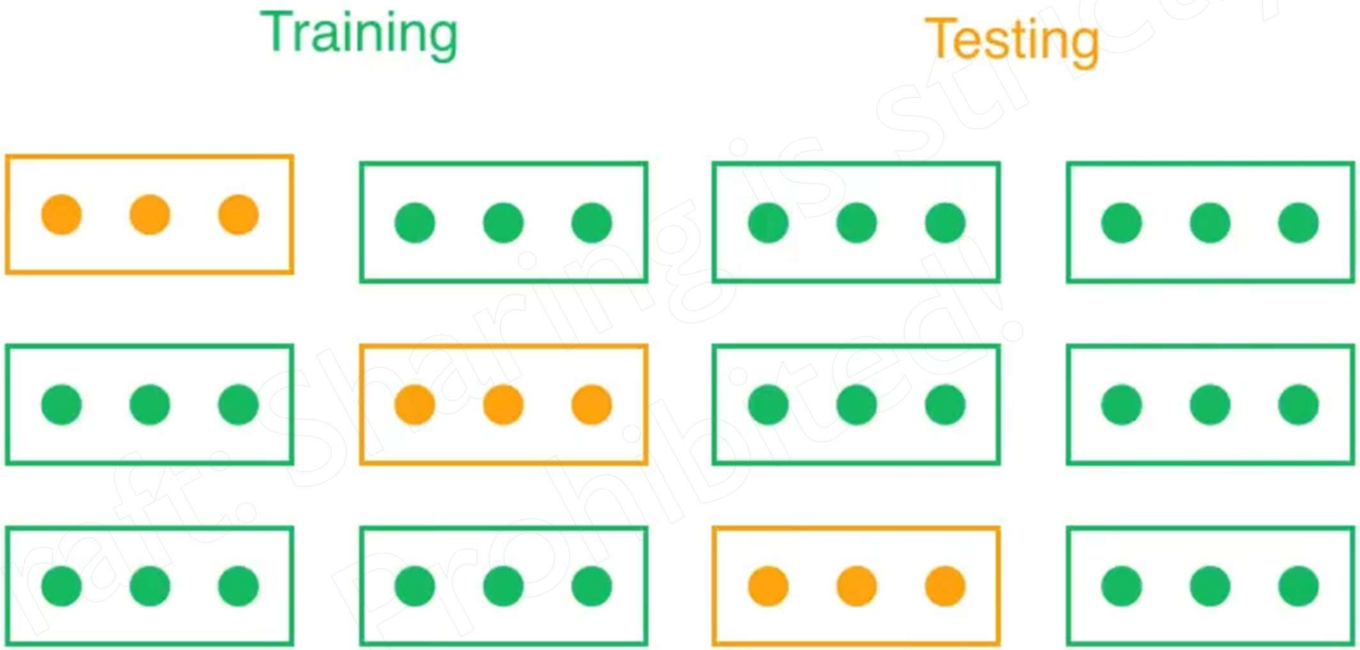
K-Fold Cross Validation



K-Fold Cross Validation

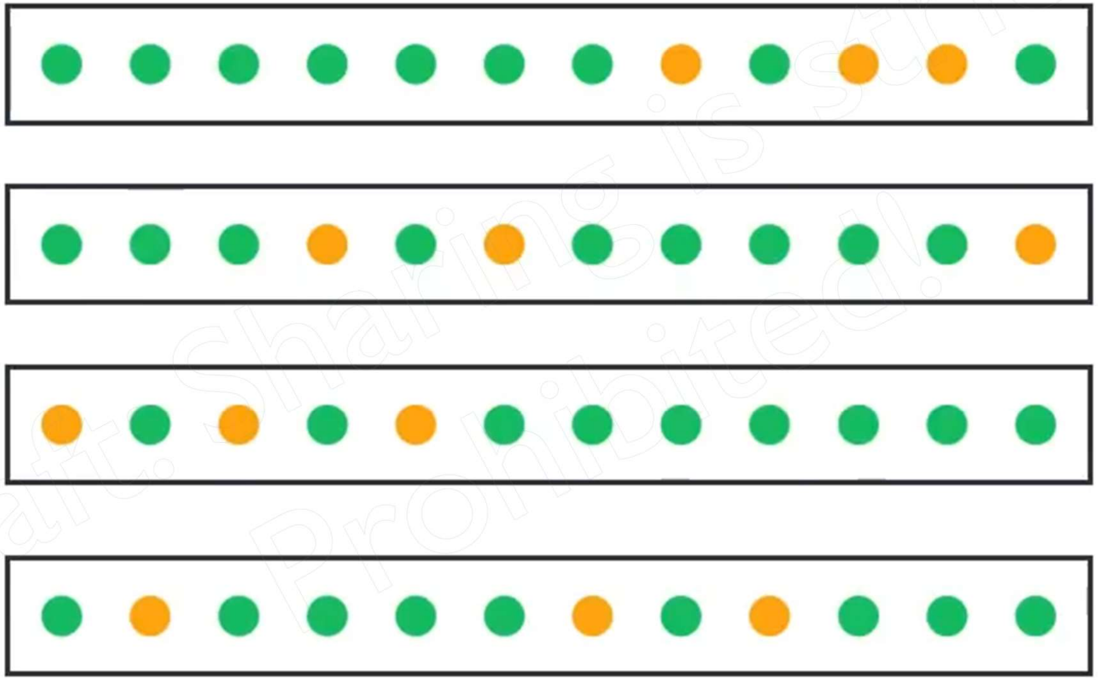


K-Fold Cross Validation



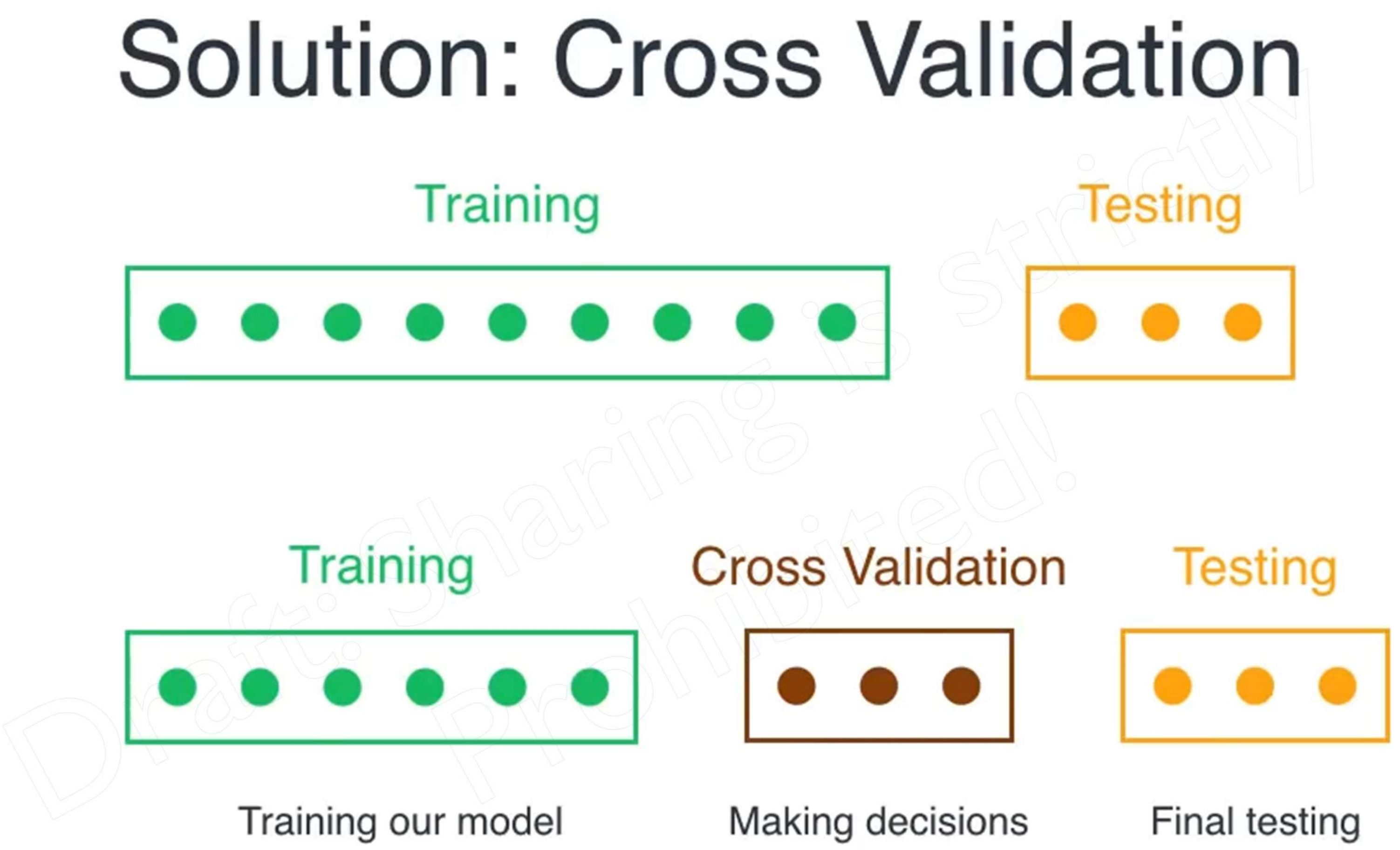
K-Fold Cross Validation

Randomizing in Cross Validation



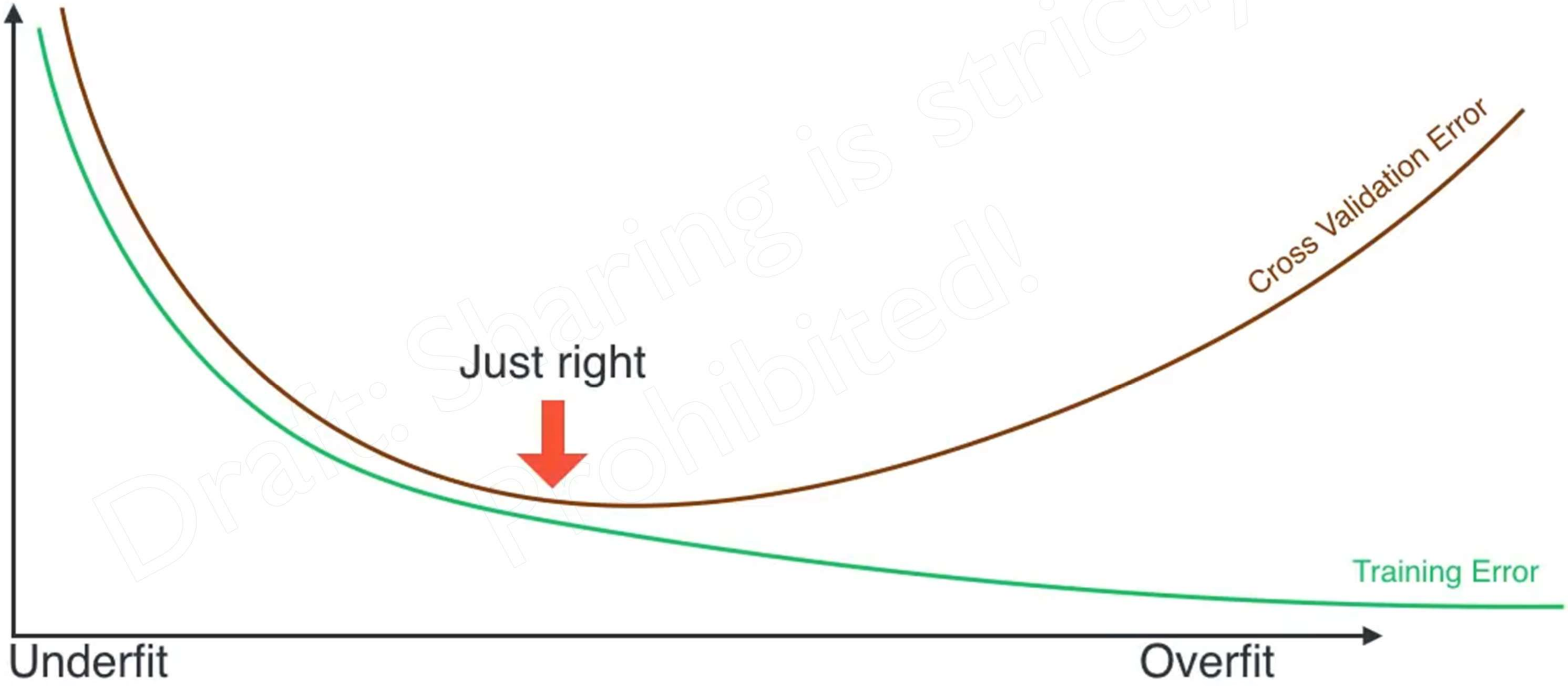
Golden Rule # 1

usually training & testing data thake, but amra jeta testing data hishebe count kori ota basically validation data.
testing data completely developer er unknown thake. so golden rules e data ke 3 ta set divide kora hoi. training data & validaiton data, jegulu deveoper ke provide kora hoi, r testing data author er kachei thake jate kaj complete houwar por authority nije check kore dekhte pare model either thik-thak kaj korche or not.



Model Complexity Graph

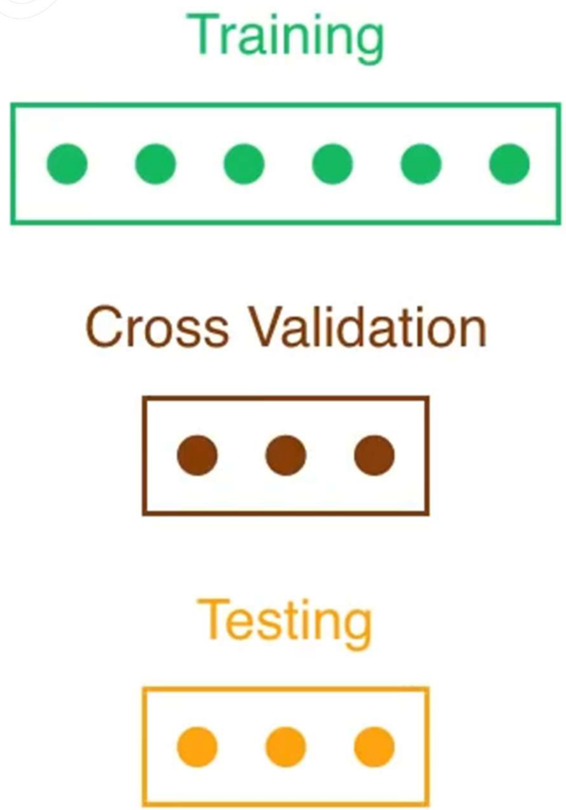
Model Complexity Graph



Training data: Train a bunch of models

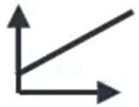
Cross validation data: Pick the best one of the models

Training a Logistic Regression Model



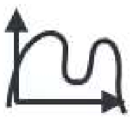
Training a Logistic Regression Model

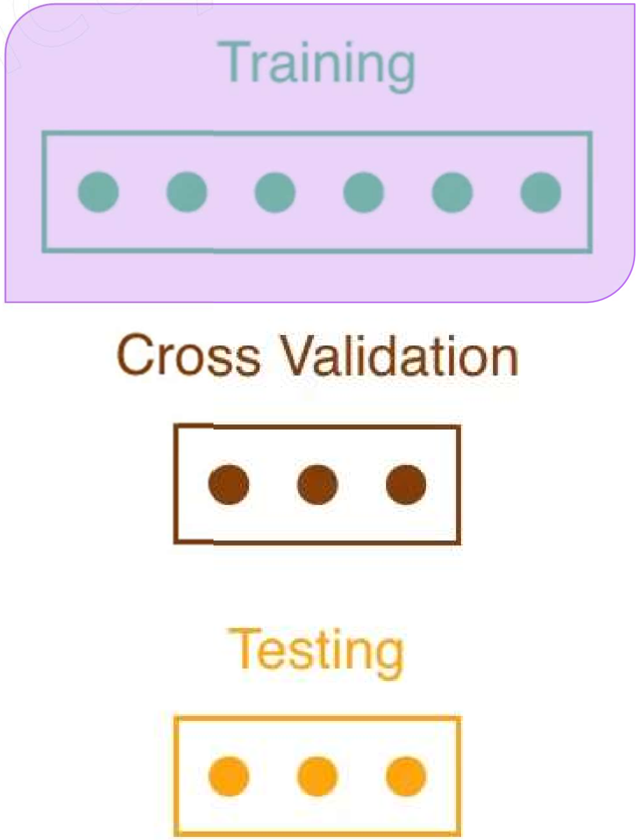
Hyperparameters Parameters

Degree = 1 

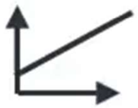
Degree = 2 

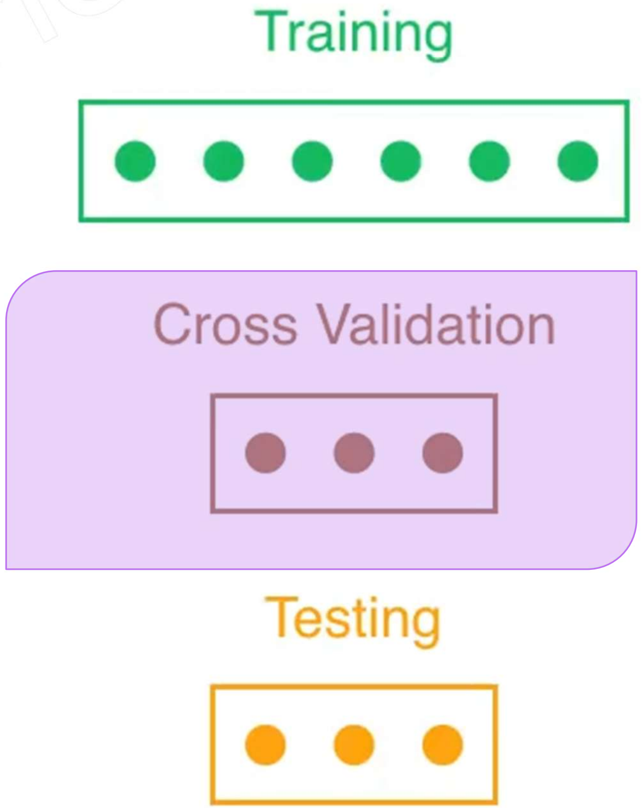
Degree = 3 

Degree = 4 



Training a Logistic Regression Model

Hyperparameters	Parameters	F1 Score
Degree = 1		0.5
Degree = 2		0.8
Degree = 3		0.4
Degree = 4		0.2



Training a Decision Tree

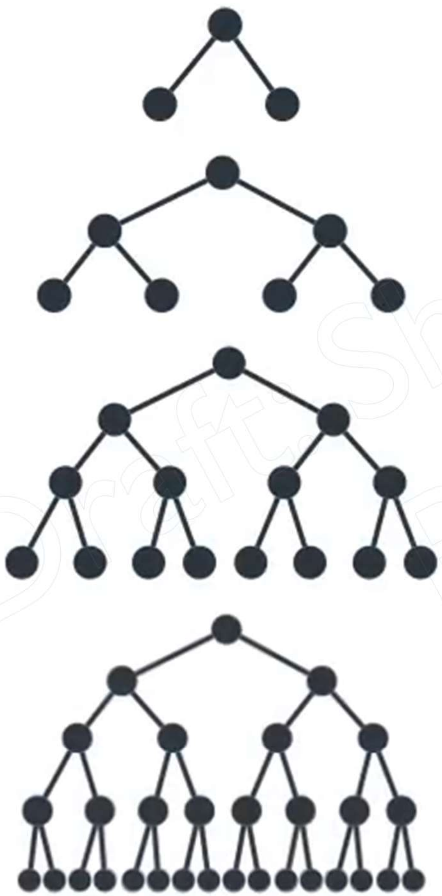
Hyperparameters Parameters

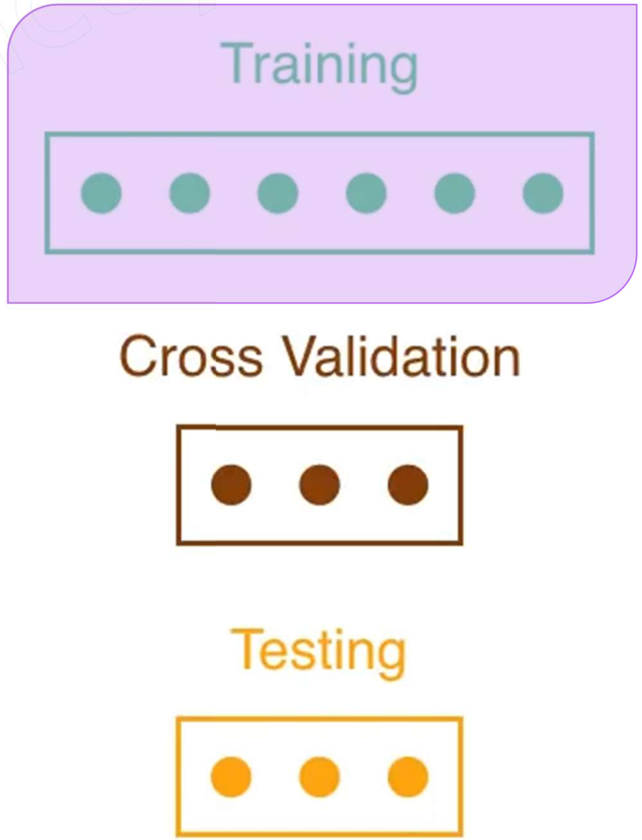
Depth = 1

Depth = 2

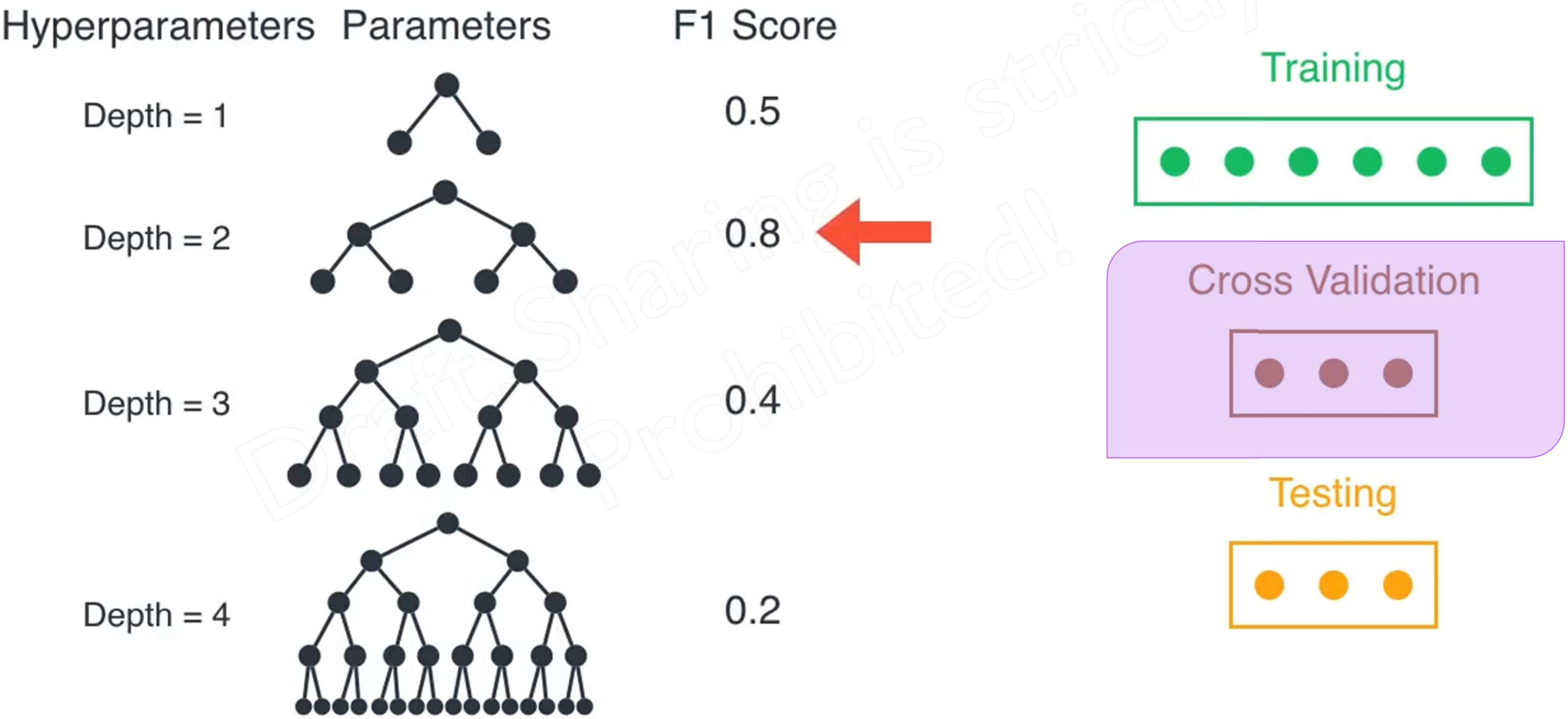
Depth = 3

Depth = 4





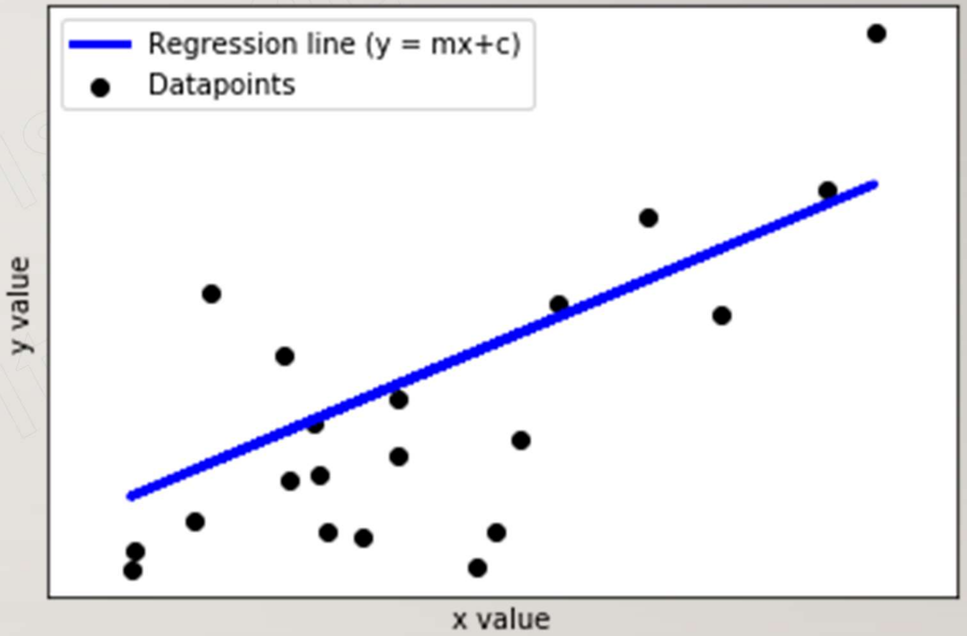
Training a Decision Tree



Parameter vs Hyperparameter

What is Model Parameter?
A model parameter is a variable whose value is estimated from the dataset. Parameters are the values learned during training from the historical data sets.

What is Hyperparameter?
A hyperparameter is a configuration variable that is external to the model. It is defined manually before the training of the model with the historical dataset. Its value cannot be evaluated from the datasets.



Parameter: m and c
Hyperparameter: Degree of polynomials, number of iterations, acceptable error thresholds, etc.

How to solve a problem

How to solve a problem

Problem

How to solve a problem

Problem

Tools

How to solve a problem

Problem

Tools

Measurement

How to solve a problem

Problem

Tools

Measurement

Measure each tool's performance

Pick the best tool

How to solve a problem

Problem

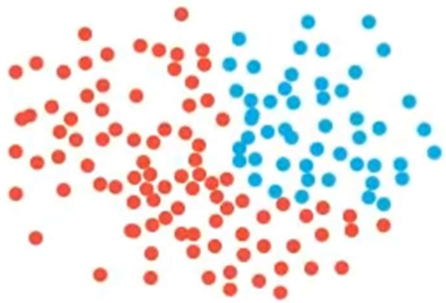
Tools

Measure each tool's performance

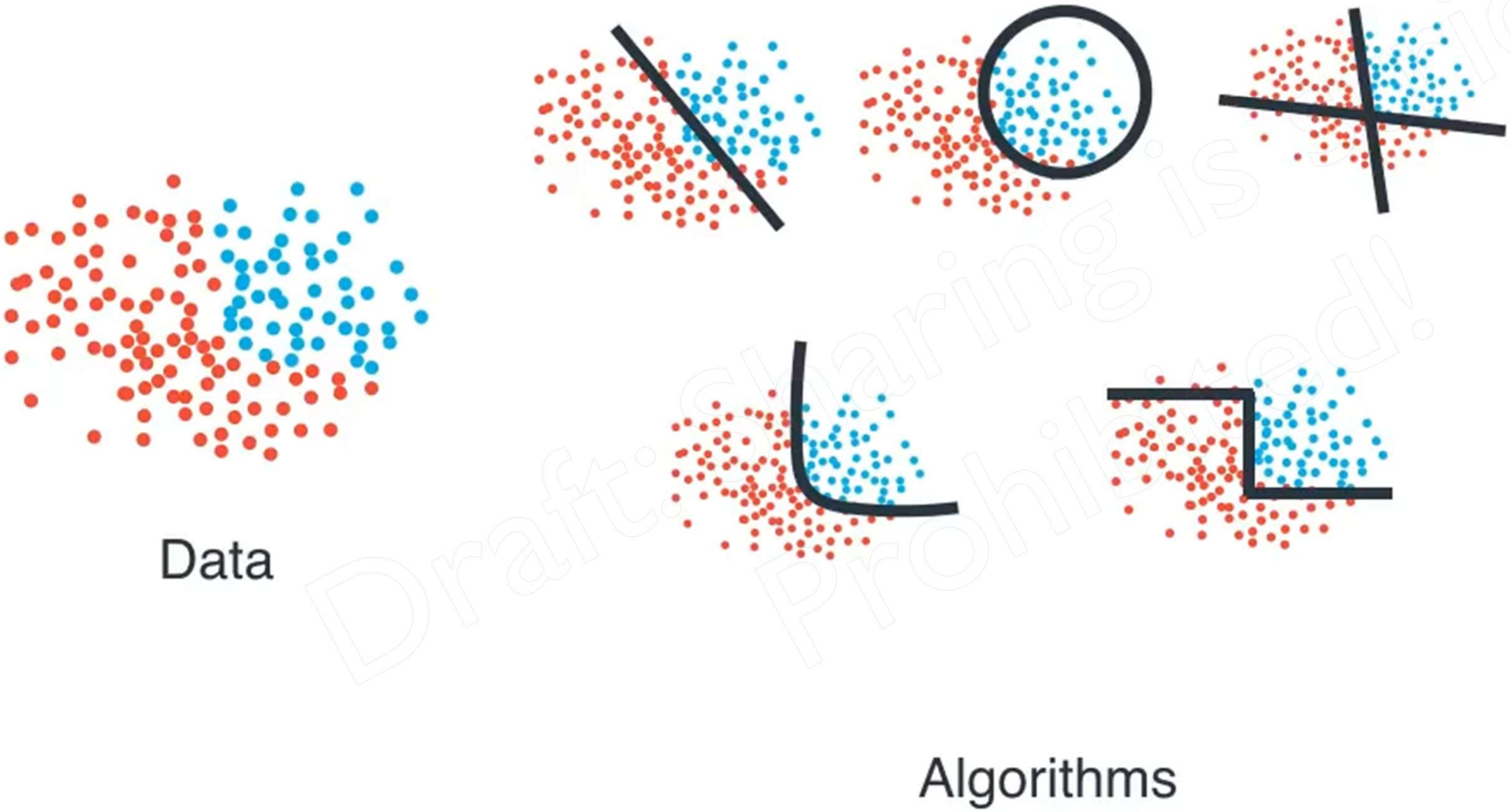
Pick the best tool

Measurement

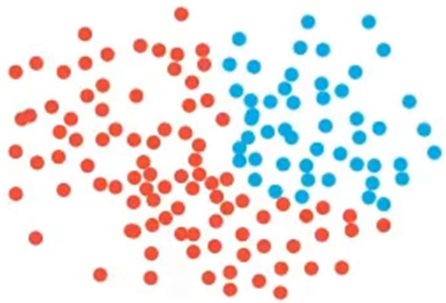
How to use machine learning



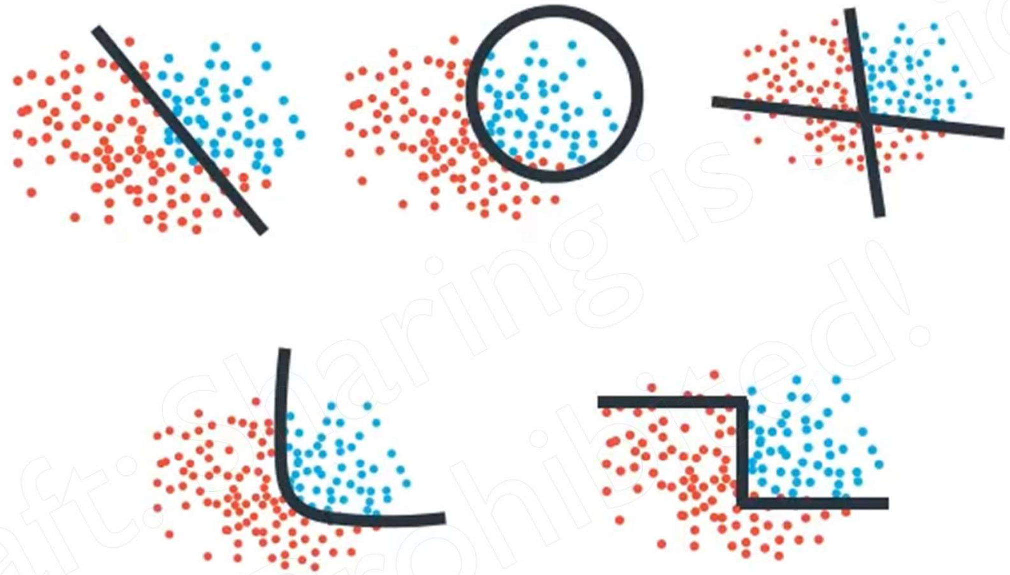
How to use machine learning



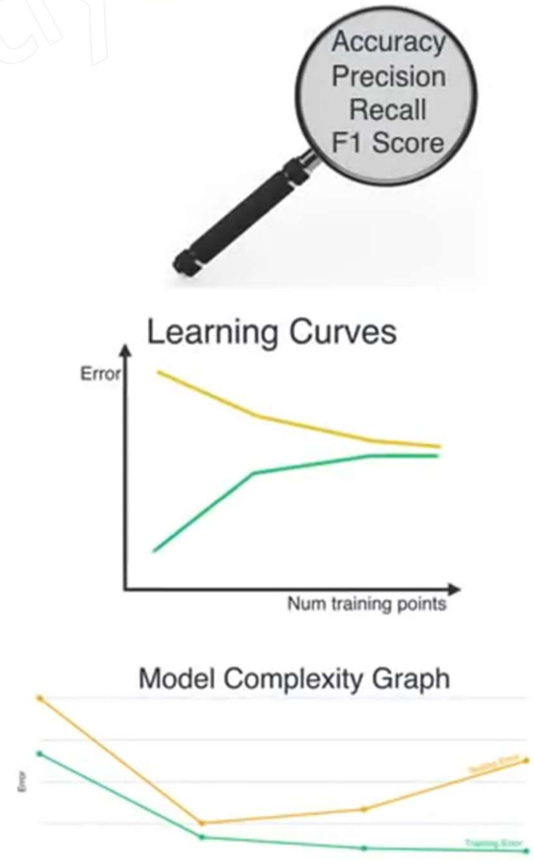
How to use machine learning



Data

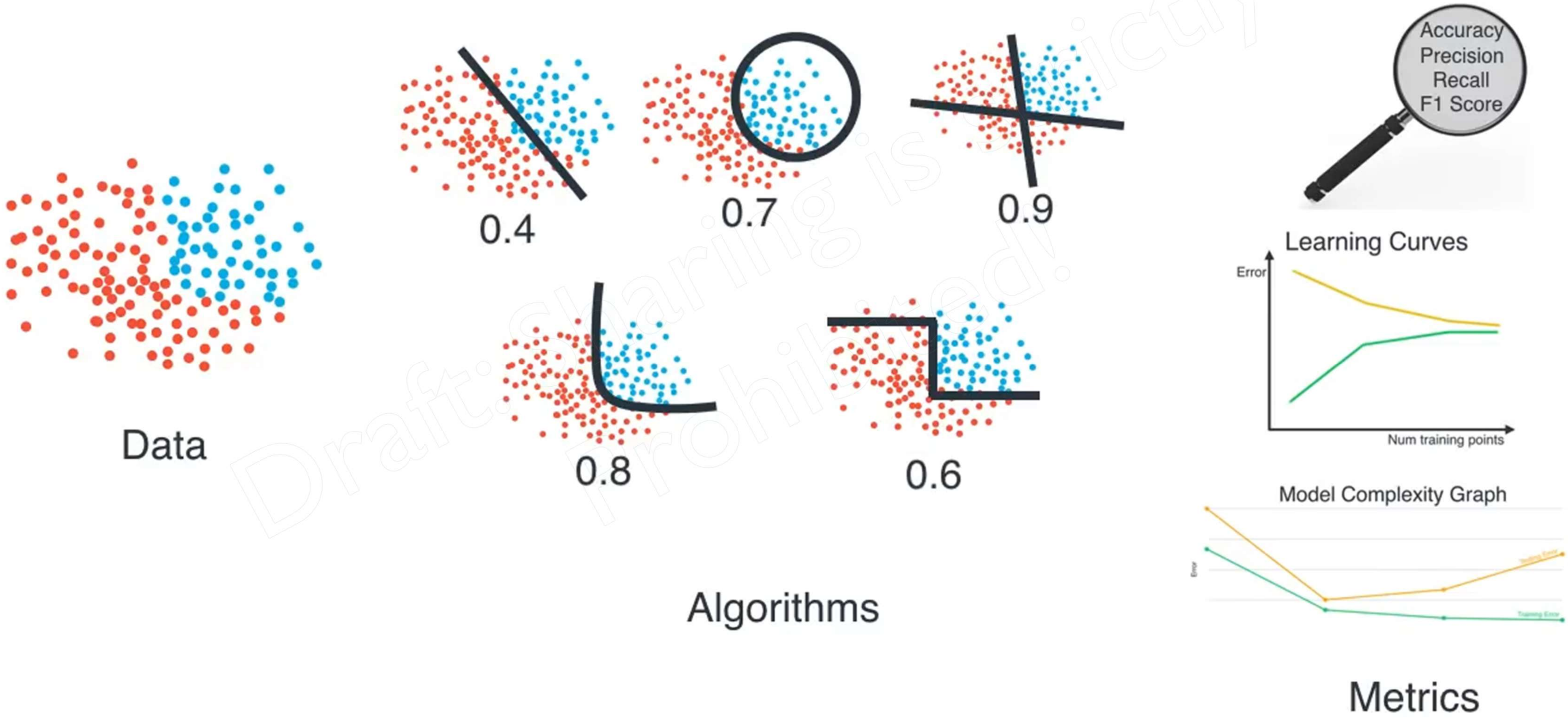


Algorithms

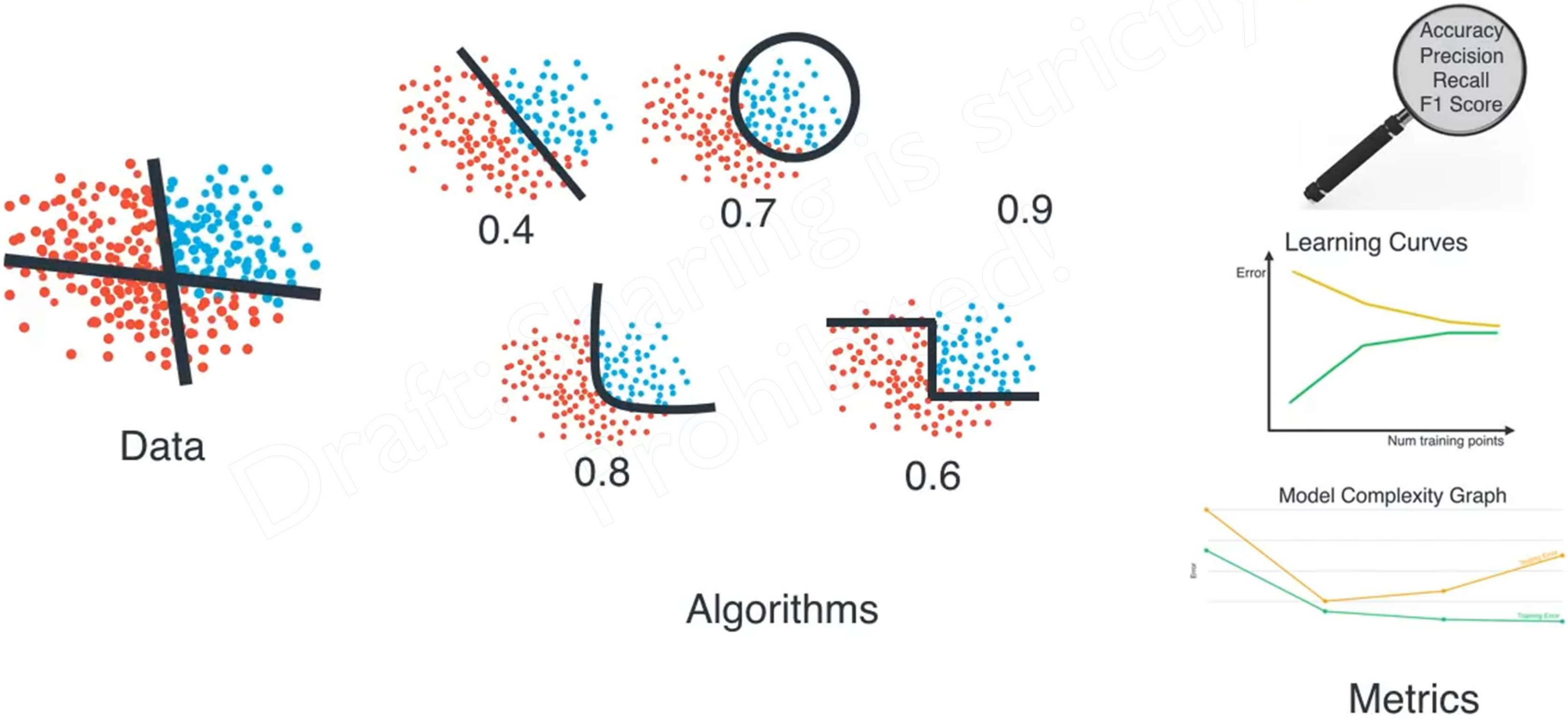


Metrics

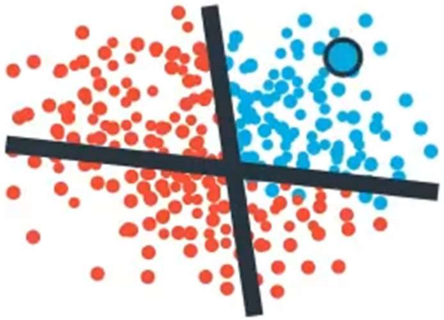
How to use machine learning



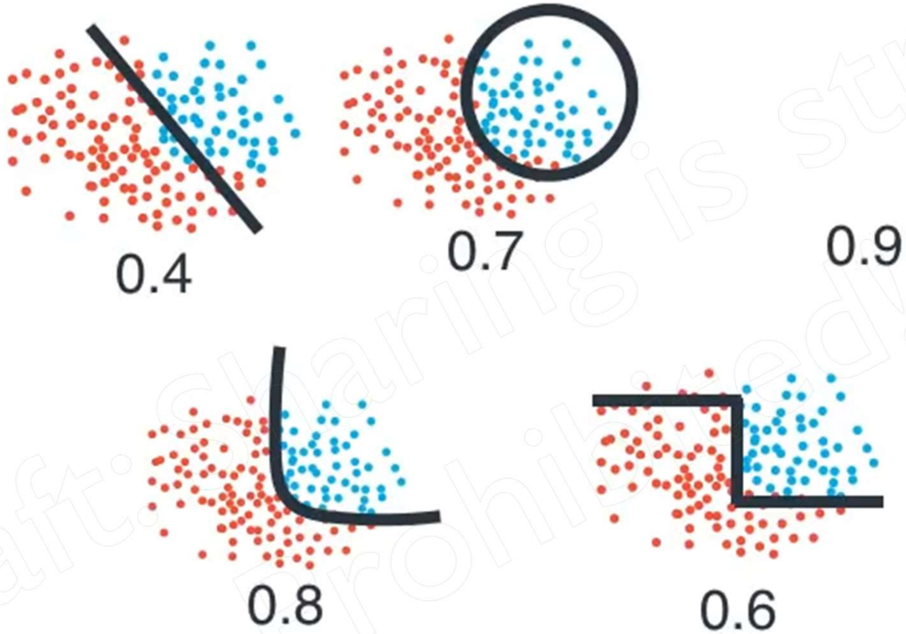
How to use machine learning



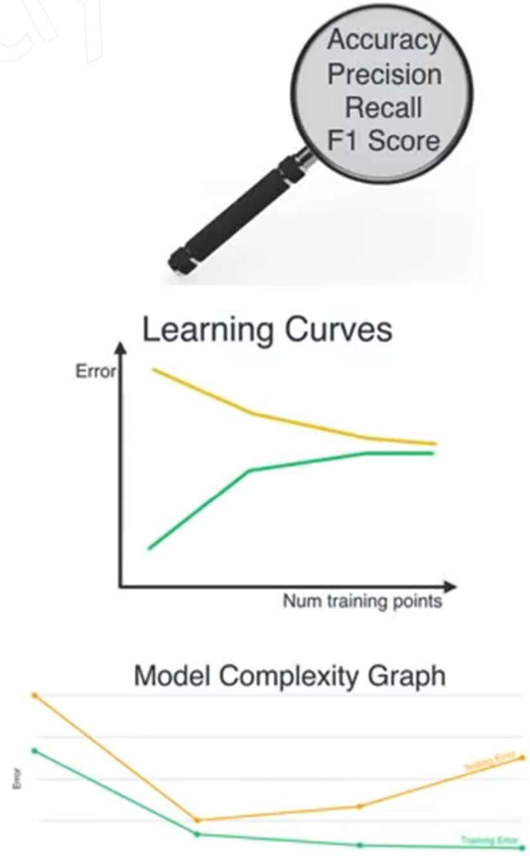
How to use machine learning



Data



Algorithms



Metrics

THANK YOU!